# Regulatory Capture via Relative Prices*

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**Abstract**

Large firms lobby for regulations that raise production costs, their own included. I show when and why this pays. The regulations I study have regressive incidence, an incidence over marginal costs that decreases with productivity. Under such a regulation every price rises, but the prices of the firms with the least incidence rise least, so they become relatively cheaper; demand reallocates toward them, and they grow, gaining revenue, profit, and market share. In a differentiated industry with homothetic single-aggregator demand, a firm gains if and only if the regulation raises the price aggregator proportionally more than its own marginal cost. Specializing to CES makes log firm size linear in log productivity, the slope being the productivity–size gradient. The change in the gradient from the regulation is the elasticity of incidence; its sign separates the types of incidence. Regressive incidence steepens the gradient, while both proportional incidence and incidence over fixed costs leave the gradient unchanged. In Nielsen scanner data, the Food Safety Modernization Act steepens the gradient in the product categories where the *Salmonella*-control duty is the firms's own and moves nothing in paired placebo categories; in oil-roasted nuts, two firms selling ten-to-one before the act ended the post period selling nearly thirty-to-one. The largest firms in the affected categories lobbied for the act. In general equilibrium, with inelastic demand across industries, the mechanism intensifies, since the industry's higher price index draws expenditure in from the rest of the economy; the regulation is a welfare loss, with a closed form identified by the estimated elasticity, and the estimates imply category price indices up to ten percent higher with only the top fifth to third of firms gaining. A researcher modeling these differentiated markets as homogeneous would report welfare gains where losses occurred in four of the categories I study.

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# 1 Introduction

In September 2016 the largest reform of U.S. food-safety law since 1938 came into force, the Food Safety Modernization Act (FSMA). The largest food manufacturers supported the act, and lobbied for it throughout its passage: a law that raised their own compliance costs. Its justification came from the 2008–09 Peanut Corporation of America *Salmonella* outbreak, in which contaminated peanut paste shipped downstream into hundreds of products, leaving over seven hundred sickened and nine dead. Interestingly, the act would not have prevented the outbreak, as PCA already ran the monitoring FSMA made mandatory. PCA ran *Salmonella* tests, and shipped despite the tests coming back positive, under falsified certificates. Its executives were convicted under statutes that predate the act.[1] So, why would firms lobby for a rule that raises their own costs?

Amazon has asked Congress to double the federal minimum wage, and campaigned publicly for it;[2] Apple has called on Congress for stricter data-privacy regulation; Meta, for content-moderation legislation.[3] None of these regulations would be free for the firm demanding it. Amazon buys from suppliers, and outsources cleaning and delivery services to contractors that pay minimum wage; a privacy statute would restrict Apple's own data collection and usage; and a moderation mandate would subject Meta's own platforms to compliance and reporting obligations. Each case poses the same question: how does a firm benefit from a regulation that makes production more expensive?

[1]PCA's Blakely, Georgia plant carried a "superior" rating from its private auditor in the year before the outbreak, and records produced at trial showed lots shipped after positive *Salmonella* results on at least a dozen occasions in 2007–08, under falsified certificates of analysis. Its executives were convicted in 2014 of fraud, conspiracy, and the introduction of adulterated food into interstate commerce, provisions of the Food, Drug, and Cosmetic Act and the criminal code.

[2]Amazon raised its own minimum wage to $15 an hour on October 2, 2018, jointly with the pledge to lobby Congress to double the federal minimum, then $7.25. But, the $15 internal floor was already the effective wage for many of Amazon's full-time workers: the announced raise arrived bundled with the phase-out of restricted-stock grants and monthly attendance bonuses averaging $1,800–$3,000 a year; see, e.g., "Despite minimum wage increase, some Amazon workers say losing stock options and bonuses means they will make less," *TechCrunch*, October 3, 2018. As part of the public campaign, Bezos closed his letter to shareholders with a dare, "Today I challenge our top retail competitors (you know who you are!) to match our employee benefits and our $15 minimum wage. Do it! Better yet, go to $16 and throw the gauntlet back at us." (Amazon, 2018 Letter to Shareholders, April 2019.)

[3]Apple endorsed "comprehensive federal privacy legislation" before the Senate Commerce Committee (Bud Tribble, hearing on Examining Safeguards for Consumer Data Privacy, September 26, 2018); Tim Cook repeated the call to Congress in "It's Time for Action on Privacy" (*TIME*, January 2019), and to regulators worldwide in his keynote to the International Conference of Data Protection and Privacy Commissioners (Brussels, October 2018). Zuckerberg first called for content rules in "The Internet needs new rules" (*Washington Post*, March 30, 2019), then took the request to Congress twice: he told the Senate Commerce Committee that Congress should update Section 230 (October 28, 2020, at a hearing devoted to the law), and before the House Energy and Commerce Committee (March 25, 2021) he proposed conditioning Section 230 liability protection on platforms demonstrating systems to identify and remove unlawful content.

The regulations these firms asked for have an asymmetric incidence over the firms they would bind. FSMA, for example, was far cheaper for its supporters, the largest manufacturers, since the text that passed largely codifies practices (hazard analysis, environmental monitoring, hygienic zoning) they had implemented since the 1990s, mainly because exporting to the EU and Japan required them; for smaller rivals, the same obligations meant new programs and new personnel. Amazon's exposure to the federal floor would be only indirect, through its suppliers and outsourced services, since its own workers already earn above the proposed minimum. Apple's revenue, unlike that of rivals such as Android, does not rely on the targeted advertising a privacy statute would restrict.[4] And Meta already runs the industry's largest content-moderation infrastructure, while smaller platforms would have to build one.[5]

The answer this paper develops is that asymmetric incidence over marginal cost is enough for a firm to profit from a regulation, and the proposed mechanism works through *relative prices*. When a regulation raises every firm's cost, every price rises — but the prices of the firms with the least incidence rise least. Despite becoming more expensive, they become relatively cheaper: demand reallocates toward them, and they grow (gain revenue, profit, and market share) while their rivals shrink. The regulations I focus on are the ones whose incidence over the marginal cost decreases with productivity, the ones with *regressive incidence*, e.g., regulations where the most productive firms already run the required systems, use less of the regulated input, or pay above the proposed floor.

Despite being intuitive, this mechanism has not been studied in the literature. In every paper I could find, marginal-cost-raising regulation with asymmetric incidence is studied through the homogeneous-good lens of *raising rivals' costs*, in partial equilibrium (Salop and Scheffman, 1983; Fowlie et al., 2016); and when regulation with asymmetric impact is studied in differentiated goods, it affects only fixed costs (Rebeyrol, 2023; Macedoni and Weinberger, 2026). In neither setting do relative prices move; a homogeneous good has a single price, and selection through fixed costs leaves relative prices among survivors unchanged. In one sentence, this paper is *raising rivals' costs* in differentiated goods, brought to general equilibrium.

Within industries, the distribution of size across firms has grown steadily more unequal for at least a century (Kwon et al., 2024), with important implications. For example, labor's declining share is driven by the reallocation of sales toward each industry's largest, low-labor-share firms (Autor et al., 2020). The aggregate markup has risen since 1980 because within-

[4] Android is licensed for free and is funded by targeted advertising that uses the data it collects.

[5] Some 30,000 people on safety and security, roughly half reviewing content, by the company's own count (Zuckerberg, "A Blueprint for Content Governance and Enforcement," November 2018).

industry market share reallocated toward larger, high-markup firms (De Loecker et al., 2020); with the resulting markup dispersion lowering aggregate productivity and welfare (Edmond et al., 2023). A fatter-tailed size distribution makes the overall economy less stable as idiosyncratic shocks to large firms do not average out: shocks to the hundred largest U.S. firms account for one third of output-growth fluctuations (Gabaix, 2011). Finally, inequality between workers is (mostly) inequality between firms, as two thirds of the rise in U.S. earnings dispersion since the late 1970s occurred *between* firms rather than within them (Song et al., 2019). None of these literatures is about regulation, but each is a reason to care about the regulations that can reshape the size distribution they study.

Section 2 develops the mechanism in partial equilibrium, for a differentiated industry whose demand is homothetic with a single price aggregator (Matsuyama and Ushchev, 2017, 2020). The main theorem states that a firm grows (gains revenue, profit, and market share) whenever the price aggregator rises proportionally more than the firm's own marginal cost. In a regressive incidence regulation, this implies that the most productive firms grow, the least productive firms shrink, and the overall size distribution becomes more unequal. To set up the estimating equation for the empirical section, I specialize the setting to CES preferences. Under CES, log firm size is linear in log productivity; I call the slope the *productivity–size gradient* (how steeply firm size rises with productivity). A regulation with regressive incidence over marginal costs steepens the gradient; progressive incidence flattens it; proportional incidence and incidence over fixed costs leave it unchanged. To estimate the change in the gradient and test for the type of incidence I use Poisson pseudo-maximum-likelihood, since log-linearity implies that the estimating equation for incidence is exponential. I also show that the estimation, as proposed, is independent of any general-equilibrium effects.

Section 3 takes the estimation to FSMA using Nielsen scanner data, 2012–2019, around the September 2016 compliance date, in the product classes exposed to *Salmonella*, the pathogen behind the outbreak that justified the reform: dry, ready-to-eat foods, long treated as low-risk because they are dry, and brought under mandatory preventive controls only afterward. The design leans on contrasts *within* product categories: same shelves, similar demand shocks, but different regulatory burden. Within a category, FSMA binds where *Salmonella* control is the facility's own, non-offloadable duty; similar products who are exempt supply the placebos.

I find that the gradient steepens (i.e., incidence over marginal costs is regressive) in the categories where FSMA binds, and it remains unchanged in the paired placebos. Among nuts, the gradient rose by 0.24 for dry roasters and by 0.46 for oil roasters, who must control recontamination in their own plants. The effect in oil-roasted nuts means that two firms selling ten-to-one in 2011 ended the post period selling nearly thirty-to-one. For handlers of raw nuts, whose monitoring duty is offloadable to their supplier, the gradient did not move. Among dried fruit, it rose for true dried fruit and stayed flat for fruit snacks, whose terminal cook means that nothing needs monitoring. Among chocolate, it rose where nut and peanut-butter fillings enter after sterilization, so the facility is the last line of control, and stayed flat for plain bars who offload the duty to cocoa roasters. I also study categories that offer no paired placebo. In those, the gradient rose as well: by 0.10 in peanut butter, the category whose outbreak justified the statute; by 0.06 in breakfast cereal; and in seasoning and in flavored crackers. The top firms in these categories lobbied for FSMA. The Grocery Manufacturers Association (GMA), the trade association of the largest food manufacturers, supported the act and filed forty disclosures on it; the largest almond processor filed fifty, the peanut shellers' association forty-one, and the cereal and chocolate majors dozens more; and the very smallest producers (those with $\leq$ \$500$k$/yr in sales) fought their way to an exemption, claiming compliance would cost them more than half their profits.

Section 4 brings the mechanism to general equilibrium in a nested CES economy in which a single industry receives a regulation with regressive incidence, and shows that the top firms of that industry grow (gain revenue, profit, and market share) under the standard assumption that demand across industries is inelastic (Atkeson and Burstein, 2008; Comin et al., 2021; Andreyeva et al., 2010). The same section takes the model structurally to measure the welfare cost of FSMA. With Pareto-distributed marginal costs, the estimated change in the gradient is equivalent to a shift of the distribution's shape parameter, fattening the tail of the marginal cost distribution and concentrating the industry. Calibration uses the shape parameter estimated from the pre-period size distribution and demand elasticities from published estimates (Hottman et al., 2016); the welfare cost then has a closed form identified by the gradient estimate itself. Across the treated categories, the estimated gradient changes imply that category price indices rise by up to ten percent, that only the top fifth to third of firms profit and grow, and that the welfare cost sums to at least \$347 million a year.

Section 5 makes the observation that every paper I could find from this literature, in the marginal-cost-homogeneous and the fixed-cost-differentiated strands alike, describes firms through an affine cost function, and models regulation as a deformation of the marginal-cost distribution or a shift in the fixed cost. I use that observation to organize the literature, and place this paper within it. Finally, Section 6 takes the simplest raising-rivals'-costs model to show the risk of studying a differentiated market through the homogeneous-good lens. A researcher modeling a differentiated market as homogeneous could wrongly conclude that a regulation with regressive incidence improves welfare: past a closed-form threshold of the leader's market share, the homogeneous model reports gains. The leader crosses that threshold in four of the products I study.

The paper proceeds as follows. Section 2 outlines the mechanism in partial equilibrium and derives the gradient-change estimating equation from a CES specialization. Section 3 presents the application. Section 4 closes the economy in general equilibrium and builds the structural quantification. Section 5 organizes the literature in cost-function space, and Section 6 asks what a researcher using the usual homogeneous-good setting would conclude. Section 7 concludes.

# 2 Regressive Incidence in Partial Equilibrium

To showcase the mechanism we start in partial equilibrium, where expenditure and input costs are given (Section 4 endogenizes them). Section 2.1 sets the environment. I show that in a differentiated industry whose demand is homothetic with a single price aggregator, the firm's problem can be rewritten in terms of relative costs and prices, so that a firm's outcomes (revenue, profit, and market share) depend only on its marginal cost relative to the price aggregator. Section 2.2 states the main theorem, a firm grows if and only if a regulation raises the aggregator proportionally more than its own marginal cost, and derives from it the regressive-incidence proposition; Section 2.3 covers the other marginal-cost deformations (proportional and progressive incidence). Sections 2.4–2.6 specialize to CES preferences, where log-linearity gives us the productivity–size gradient, and I use the changes in that gradient to estimate the elasticity of incidence and tell the competing types of incidence apart.

## 2.1 Environment

Consider a static industry with a unit continuum of differentiated varieties indexed by $\omega \in (0,1)$, each produced by a single firm. Preferences over varieties are homothetic with a single aggregator (HSA), as in Matsuyama and Ushchev (2017, 2020). Demand for variety $\omega$ is

$$x_\omega \;=\; \frac{Y}{p_\omega}\, s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right), \tag{1}$$

where $Y > 0$ is the fixed industry expenditure, $s_\omega(\cdot)$ is the share function, and $P(\mathbf{p})$ is the price aggregator,[6] a linear homogeneous function of the price profile defined implicitly by the adding-up condition

$$\int_0^1 s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right) d\omega \;=\; 1\,. \tag{2}$$

[6] $P(\mathbf{p})$ is Matsuyama and Ushchev's common price aggregator $A(\mathbf{p})$, the single statistic through which rivals' prices enter each firm's demand.

I assume throughout that the goods are gross substitutes, $s'_\omega(z) < 0$, i.e., a firm that raises its relative price loses expenditure share (under CES, $\sigma > 1$). The adding-up condition (2) then pins the aggregator down and it gives us three properties. First, $P(\mathbf{p})$ exists and is unique (Matsuyama and Ushchev, 2017). Second, $P$ is linearly homogeneous: scaling every price and $P$ by the same factor leaves every relative price (and the integral) unchanged, so by uniqueness $P(\lambda\mathbf{p}) = \lambda\, P(\mathbf{p})$. Third, $P$ is increasing in the price profile.

Firm $\omega$ has constant marginal cost $\psi_\omega > 0$ and chooses its price to maximize

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y\, s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right). \tag{3}$$

The problem can be restated in relative terms: since $\psi_\omega/p_\omega = (\psi_\omega/P(\mathbf{p}))/(p_\omega/P(\mathbf{p}))$, writing $z_\omega \equiv p_\omega/P(\mathbf{p})$ for the relative price gives

$$\pi_\omega \;=\; \left(1 - \frac{\psi_\omega/P(\mathbf{p})}{z_\omega}\right) Y\, s_\omega(z_\omega), \tag{4}$$

so maximizing over price levels is equivalent to maximizing over relative prices, and the firm's own marginal cost enters the problem only through the *relative marginal cost* $\psi_\omega/P(\mathbf{p})$. Each firm maximizes (4) over its relative price $z_\omega$, taking the aggregator as given, and trades a wider margin against a smaller share. The first-order condition is then:

$$\frac{\psi_\omega/P(\mathbf{p})}{z_\omega^2}\, s_\omega(z_\omega) \;+\; \left(1 - \frac{\psi_\omega/P(\mathbf{p})}{z_\omega}\right) s'_\omega(z_\omega) \;=\; 0. \tag{5}$$

The first term is the value of widening the margin, the second the share the wider margin gives up. Write $\zeta_\omega(z) \equiv -\,z\, s'_\omega(z)/s_\omega(z) > 0$ for the elasticity of the share function. Since demand (1) is expenditure over the own price times the share, the demand elasticity the firm perceives is $1 + \zeta_\omega(z)$, greater than one in the gross substitutes case. Multiplying (5) by $z_\omega/s_\omega(z_\omega)$ and collecting terms, the optimum satisfies

$$p^*_\omega \;=\; \mu_\omega\, \psi_\omega, \qquad \mu_\omega \;=\; \frac{1 + \zeta_\omega(z^*_\omega)}{\zeta_\omega(z^*_\omega)} \;>\; 1, \tag{6}$$

a finite markup over marginal cost (in the CES case, $\zeta = \sigma - 1$ for every variety, so $\mu = \sigma/(\sigma-1)$ for every firm). At the equilibrium profile $\mathbf{p}^* = \boldsymbol{\mu}\boldsymbol{\psi}$, profit depends on the firm's own cost only through $\psi_\omega/P$. Differentiating (4) with respect to this relative marginal cost, the envelope theorem gives

$$\frac{\partial \pi^*_\omega}{\partial\big(\psi_\omega/P\big)} \;=\; -\,\frac{Y\, s_\omega(z^*_\omega)}{z^*_\omega} \;<\; 0, \tag{7}$$

and the equilibrium revenue and market share are likewise strictly decreasing in $\psi_\omega/P$, since the optimal relative price $z^*_\omega$ rises with relative marginal cost.[7] Relative marginal cost is then the sufficient statistic for a firm's competitive position; it determines its (relative) price, its market share, its revenue, and its profit.

Without loss of generality, marginal costs are ordered by rank: $\psi_\omega = G^{-1}(\omega)$ for a continuous, strictly increasing cost distribution $G$ on $[\psi_0, \infty)$, $\psi_0 > 0$, so that low $\omega$ means low cost, i.e., the most productive firms sit near $\omega = 0$.

## 2.2 Regulations as deformations

**Definition 1** (Regulation)**.** *A* regulation *is a change in the marginal-cost profile from $\psi$ to $\psi^R$. It is* cost-increasing *if $\psi^R_\omega \geq \psi_\omega$ for all $\omega$, with strict inequality on a set of positive measure.*

We next ask which cost-increasing regulations make some firms grow.

**Theorem 1** (Who wins)**.** *Under a deformation $\psi \to \psi^R$, firm $\omega$ attains strictly higher equilibrium revenue, profit, and market share if and only if its relative marginal cost falls,*

$$\frac{\psi^R_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})},$$

*strictly lower if the inequality is reversed, and unchanged under equality.*

Proof in Appendix A.

In plain words, a firm grows when the regulation raises the price environment it competes against proportionally more than the firm's own marginal cost. Regressive incidence is then a sufficient condition for the most productive firms to grow:

**Proposition 1** (Regressive incidence)**.** *Take a cost-increasing regulation and write its incidence as $1 + \varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega$. Suppose the incidence is* regressive*: $\varepsilon_\omega \leq \varepsilon_{\omega'}$ whenever $\omega < \omega'$, with strict inequality for some pair. Then there exist $0 < \bar{\omega} \leq \bar{\omega}' < 1$ such that all firms in $(0, \bar{\omega})$, i.e., the most productive, strictly gain revenue, profit, and market share, and every firm in $(\bar{\omega}', 1)$ strictly loses.*

No firm needs to be fully exempt; it is sufficient that the most productive firms be the *least* burdened, e.g., because they already run the required systems, use less of the regulated input,

[7] Derivatives of equilibrium revenue and market share with respect to relative marginal cost: $\dfrac{\partial\,[Y s_\omega(z^*_\omega)]}{\partial(\psi_\omega/P)} = Y s'_\omega(z^*_\omega)\,\dfrac{\partial z^*_\omega}{\partial(\psi_\omega/P)} < 0$, and $\dfrac{\partial\, s_\omega(z^*_\omega)}{\partial(\psi_\omega/P)} = s'_\omega(z^*_\omega)\,\dfrac{\partial z^*_\omega}{\partial(\psi_\omega/P)} < 0$.

or pay near the proposed floor. The proof uses two properties of the aggregator from Section 2.1. Under regressive incidence every marginal cost rises at least by the most productive firms' incidence, and some rise strictly more; monotonicity and linear homogeneity then force the aggregator up by strictly more than the lowest incidence, so the most productive firms' relative marginal cost falls and Theorem 1 implies they grow. The opposite happens at the other end of the productivity distribution, every cost rises at most by the least productive firms' incidence and some by strictly less, the aggregator rises by less than it, and the least productive firms' relative costs rise. Proof in Appendix A.[8] The following corollary shows that adding selection through exit only amplifies the gains of the most productive firms.

**Corollary 1** (Exit amplifies)**.** *Suppose, in addition to the conditions of Proposition 1, that firms must pay a fixed operating cost $F > 0$ to remain active, and that a positive measure of firms exits under the regulation. Then the winning firms in $(0, \bar{\omega})$ that survive gain strictly more than in the no-exit case.*

With fewer active varieties, the adding-up condition (2) must renormalize shares over a smaller set, which pushes the aggregator further up, the same direction that made the most productive firms win. The survivors' gain is net of $F$, which the regulation leaves unchanged. Proof in Appendix A.

## 2.3 Proportional and progressive incidence

**Proposition 2** (Proportional increases are neutral)**.** *Let $\psi^R_\omega = (1+\varepsilon)\,\psi_\omega$ for all $\omega$ and some $\varepsilon > 0$. Then every firm's relative marginal cost is unchanged,*

$$\frac{\psi^R_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} = \frac{(1+\varepsilon)\psi_\omega}{P((1+\varepsilon)\boldsymbol{\mu}\boldsymbol{\psi})} = \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \qquad \textit{for every } \omega,$$

*and so are every firm's revenue, profit, and market share.*

An ad valorem tax and a uniform increase in the price of a common input bundle are proportional increases. Each scales every marginal cost by the same factor, and by linear homogeneity the aggregator scales with them, so no relative marginal cost moves. Every paper I could find on marginal-cost regulation in differentiated goods concludes that marginal-cost-increasing regulations are neutral (Macedoni and Weinberger, 2026; Anouliès, 2017); seen through Proposition 2, the neutrality is a property of the deformation those papers consider, a proportional increase in a homothetic environment, and not of marginal-cost regulation as such. Proof in Appendix A.

[8] Appendix B has two stronger variants of Proposition 1, one where incidence vanishes to zero at the top of the productivity distribution, and one that takes the form of a kink that spares the top.

**Proposition 3** (Progressive incidence)**.** *Take a cost-increasing regulation and write its incidence as* $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega$*. Suppose the incidence is* progressive*:* $\varepsilon_\omega \geq \varepsilon_{\omega'}$ *whenever* $\omega < \omega'$*, with strict inequality for some pair. Then there exist* $0 < \bar{\omega} \leq \bar{\omega}' < 1$ *such that every firm in* $(0, \bar{\omega})$ *strictly loses revenue, profit, and market share, and all firms in* $(\bar{\omega}', 1)$*, i.e., the least productive, strictly grow.*

Progressive incidence redistributes *against* the most productive, shrinking them, while the least productive firms grow. Proof in Appendix A.

*Remark* 1 (Deregulation). Every statement has a symmetric counterpart for cost-*decreasing* deformations. In particular, removing a regressive incidence regulation strictly harms the most productive firms.[9]

The next three subsections set up the estimating equation that will be used in Section 3.

## 2.4 A CES specialization

I now specialize the demand system of Section 2.1 to its CES member by picking the share functions $s_\omega(z) = z^{1-\sigma}$ with $\sigma > 1$. The adding-up condition (2) becomes $\int_0^1 (p_\omega/P)^{1-\sigma}\, d\omega = 1$, and solving for the aggregator gives the CES price index,

$$P \;=\; \left(\int_0^1 p_\omega^{1-\sigma}\, d\omega\right)^{\frac{1}{1-\sigma}}. \tag{8}$$

As before, expenditure $Y$ and the price of the input bundle are held fixed (Section 4 endogenizes them). Firm $\omega$ produces at constant marginal cost

$$\psi_\omega \;=\; \frac{e}{\varphi_\omega}, \tag{9}$$

where $e$ is the price of the input bundle, common to all firms, and $\varphi_\omega$ is the firm's productivity. With the share elasticity constant at $\zeta = \sigma - 1$, every firm charges the CES markup $\mu = \sigma/(\sigma-1)$ over its marginal cost (6), and firm $\omega$'s revenue is $r_\omega = Y\, s_\omega(z_\omega) = Y\,(p_\omega/P)^{1-\sigma}$. In logs,

$$\ln r_\omega \;=\; \underbrace{\ln Y + (\sigma-1)\ln\frac{P}{\mu\, e}}_{\text{common to all firms}} \;+\; \underbrace{(\sigma-1)}_{\text{the gradient}}\;\ln\varphi_\omega\,. \tag{10}$$

[9] Appendix G asks how far Propositions 1 and 2 extend beyond HSA demand. The equivalence between maximizing over the price and over the relative price extends to the other homothetic classes of Matsuyama and Ushchev (2017); Proposition 2 holds in all of them; Proposition 1 extends under conditions the appendix states. The appendix also covers demand systems outside these classes and finds that, without homotheticity, Proposition 2 itself fails.

Log revenue is linear in log productivity. Everything common to all firms (expenditure, the price index, the price of the input bundle, the markup) sits in the intercept; the only firm-specific term is productivity. I call the slope the *productivity–size gradient*: how steeply firm size rises with productivity. Under CES it equals $\sigma - 1$, the rate at which a productivity difference becomes a size difference.

## 2.5 What a regulation does to the gradient

Let a regulation multiply firm $\omega$'s marginal cost by $1 + \varepsilon_\omega$, the firm's incidence, with $\varepsilon_\omega \geq 0$. After the regulation the firm's marginal cost is $(1 + \varepsilon_\omega)\, e/\varphi_\omega$, so it produces as if its productivity were $\varphi_\omega/(1 + \varepsilon_\omega)$. Applying equation (10) before and after the regulation,

$$\begin{aligned}
\ln r_\omega^{\text{pre}} \;&=\; \ln Y + (\sigma - 1) \ln \frac{P}{\mu\, e} + (\sigma - 1) \ln \varphi_\omega\,, \\
\ln r_\omega^{\text{post}} \;&=\; \ln Y + (\sigma - 1) \ln \frac{P^R}{\mu\, e} + (\sigma - 1) \ln \varphi_\omega \;-\; (\sigma - 1) \ln(1 + \varepsilon_\omega)\,,
\end{aligned}$$

and, with expenditure and the input price held fixed, differencing the two lines, $\Delta \ln r_\omega = \ln r_\omega^{\text{post}} - \ln r_\omega^{\text{pre}}$, leaves

$$\Delta \ln r_\omega \;=\; \underbrace{(\sigma - 1)\, \Delta \ln P}_{\text{common to all firms}} \;-\; (\sigma - 1)\; \ln(1 + \varepsilon_\omega)\,. \tag{11}$$

The price index's response is common to every firm; only the cross-firm *pattern* of incidence can move the slope.

If incidence is *proportional*, the last term of (11) is common too, so the whole effect is an intercept shift, and the gradient does not move (using Prop. 2, $\Delta \ln P = \ln(1 + \varepsilon)$ so $\Delta \ln r_\omega = 0$). If the regulation instead works through *selection* (a fixed-cost increase that forces exit at the bottom), exiters leave the economy, pushing the price aggregator up, but no surviving firm's marginal cost changes, so the gradient again does not move (still, $\Delta \ln r_\omega > 0$ for all surviving firms, same mechanism behind Cor. 1). Appendix I derives this invariance in two benchmark models, the competitive industry of Hopenhayn (1992) and the selection model of Melitz (2003). But if incidence is *regressive*, the last term of (11) shrinks with productivity, and that is enough for the most productive firms to grow (gain revenue, profit, and market share). Under CES the price index's response has a closed form,

$$\left(\frac{P^R}{P}\right)^{1-\sigma} \;=\; \int_0^1 s_\gamma\, (1 + \varepsilon_\gamma)^{1-\sigma}\, d\gamma, \tag{12}$$

with $s_\gamma$ the pre-regulation share. The index's rise is a share-weighted power mean of the incidences. To first order in the incidences the exponents cancel, and the log change is the share-weighted average of the log incidences. By (11), firm $\omega$ therefore grows if and only if

$$\ln(1+\varepsilon_\omega) \ < \ \Delta \ln P \ \simeq \ \int_0^1 s_\gamma \ \ln(1+\varepsilon_\gamma)\, d\gamma, \tag{13}$$

its own incidence below the industry's share-weighted average. With regressive incidence the inequality holds for the most productive firms, those in $(0, \bar{\omega})$, a special case of Proposition 1.

The remaining question is by how much the gradient moves. Write the *elasticity of incidence* with respect to productivity as

$$\tau \ \equiv \ -\frac{d\ln(1+\varepsilon_\omega)}{d\ln\varphi_\omega} \ > \ 0, \tag{14}$$

then the change in the gradient is $(\sigma-1)\,\tau$.[10] Figure 1 depicts how the gradient changes in response to the different types of incidence.

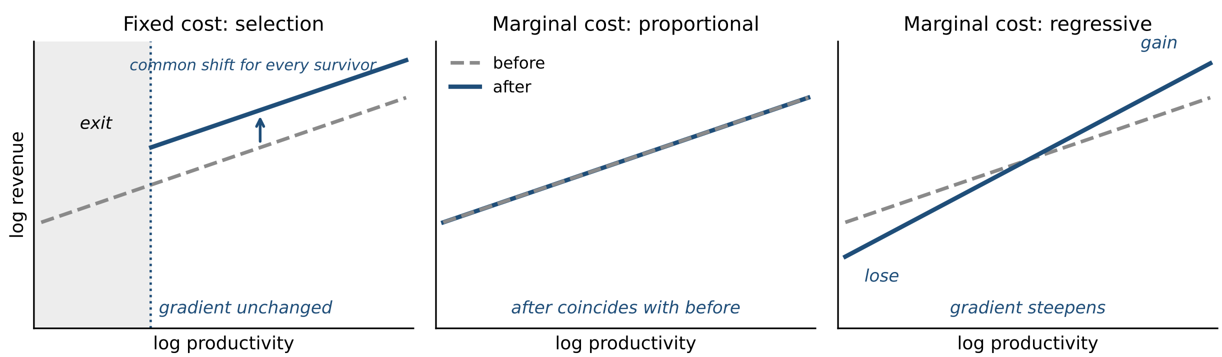


Figure 1: The productivity–size gradient under three deformations, before the regulation (dashed grey) and after (solid blue). A fixed-cost increase removes the least productive firms and shifts every survivor up by a common factor, leaving the gradient unchanged. A proportional incidence leaves every revenue unchanged. A regressive incidence steepens the gradient.

## 2.6 Measuring the change

Since I cannot observe productivity directly, I measure it with baseline revenue (a firm's size before the estimation window).[11] This is a model-correct proxy, with the added benefit that

[10] *Progressive* incidence flips the sign, $\tau < 0$.

[11] The positive relationship between firm size and productivity is a robust finding of the empirical productivity literature: aggregate productivity exceeds the unweighted firm average precisely because revenue shares and productivity are positively correlated within industries (Olley and Pakes, 1996; Syverson, 2011;

it removes the nuisance term $\sigma - 1$ from the change in the gradient at the time of estimation, so the elasticity of incidence is measured directly. By (10), log revenue at the baseline is an affine transform of log productivity,

$$\ln r_\omega^0 \;=\; a_0 \;+\; (\sigma-1)\ln\varphi_\omega \qquad\Longleftrightarrow\qquad \ln\varphi_\omega \;=\; \frac{\ln r_\omega^0 - a_0}{\sigma-1}, \tag{15}$$

where $a_0$ collects the (common) baseline intercept, so baseline revenue ranks firms exactly as productivity does, and spaces them in proportion to $\sigma - 1$. With the elasticity of incidence constant, (14) says $\ln(1+\varepsilon_\omega) = c - \tau \ln\varphi_\omega$ for some common level $c$; substituting this and (15) into (11) gives

$$\Delta \ln r_\omega \;=\; \text{const} \;+\; \tau\, \ln r_\omega^0, \tag{16}$$

where the constant collects $a_0$, $c$, and the price aggregator's response.

Write $A^{\text{pre}}$ for the common terms of (10) in the pre period. With no regulation, pre-period log revenue is (10) with productivity expressed through the baseline (15),

$$\ln r_\omega^{\text{pre}} \;=\; \bigl(A^{\text{pre}} - a_0\bigr) \;+\; \ln r_\omega^0 \;:$$

a line of unit slope in the baseline, every common term in the intercept. With the regulation, the post period adds the response (16),

$$\ln r_\omega^{\text{post}} \;=\; \ln r_\omega^{\text{pre}} + \Delta\ln r_\omega \;=\; \bigl(A^{\text{pre}} - a_0 + \text{const}\bigr) \;+\; (1+\tau)\,\ln r_\omega^0\,.$$

Let $\text{post}_t$ indicate the post period. Period revenue is then built from its two constituents,

$$\begin{aligned}\ln r_{\omega t} \;&=\; (1-\text{post}_t)\,\ln r_\omega^{\text{pre}} \;+\; \text{post}_t\,\ln r_\omega^{\text{post}}\\ &=\; \bigl(A^{\text{pre}} - a_0\bigr) \;+\; \text{const}\cdot\text{post}_t \;+\; \ln r_\omega^0 \;+\; \tau\,\bigl(\ln r_\omega^0 \times \text{post}_t\bigr).\end{aligned}$$

Grouping: the terms that vary only with the period are the time effects $\delta_t = \bigl(A^{\text{pre}} - a_0\bigr) + \text{const}\cdot\text{post}_t$; the baseline term varies only with the firm and is the firm effect $\delta_\omega = \ln r_\omega^0$. Renaming the coefficient of the interaction term $\beta$,

$$\ln r_{\omega t} \;=\; \beta\,\bigl(\ln r_\omega^0 \times \text{post}_t\bigr) \;+\; \delta_\omega \;+\; \delta_t, \qquad \beta = \tau,$$

Bartelsman et al., 2013), and the relationship has been found across seventeen countries in manufacturing and services alike (Berlingieri et al., 2018).

equivalently (and the form that we will be using),

$$r_{\omega t} \;=\; \exp\Big\{\,\beta\left(\ln r^0_\omega \times \mathrm{post}_t\right) \;+\; \delta_\omega \;+\; \delta_t\,\Big\}\,. \tag{17}$$

Any aggregate movement in expenditure $Y$ or in the price of the input bundle $e$ would be included in the time effects, even though this section held them fixed for exposition; Appendix H allows them to move, and rebuilds the derivation, arriving at the same estimating equation.

Another way to arrive at $\beta$ is from (16) directly: $\beta$ is its cross-firm slope,

$$\beta \;=\; \frac{\mathrm{Cov}\big(\Delta \ln r_\omega,\, \ln r^0_\omega\big)}{\mathrm{Var}\big(\ln r^0_\omega\big)} \;=\; \frac{(\sigma-1)\,\tau\;\cdot\;(\sigma-1)\,\mathrm{Var}(\ln\varphi_\omega)}{(\sigma-1)^2\,\mathrm{Var}(\ln\varphi_\omega)} \;=\; \tau\,.$$

We can use the sign of $\beta$ to test the type of incidence:

$$\text{regressive incidence} \;\Rightarrow\; \beta>0 \qquad \text{progressive incidence} \;\Rightarrow\; \beta<0$$
$$\text{proportional incidence, selection through fixed costs} \;\Rightarrow\; \beta=0.$$

The incidence test is not special to CES; what CES brings is the linearity that makes $\beta$ the elasticity $\tau$ itself.

# 3 The Application: FSMA

Next, I estimate the elasticity of incidence of the Food Safety Modernization Act.

## 3.1 The regulation

The Food Safety Modernization Act (signed January 2011, finalized September 2015) is the largest reform of U.S. food-safety law since 1938. Its justification came from the 2008–09 Peanut Corporation of America outbreak, in which *Salmonella*-contaminated peanut paste travelled downstream into hundreds of products, sickened over seven hundred people, killed nine, and ended in criminal convictions. Yet the regulation that resulted would not have prevented it, as PCA already ran the monitoring the statute later made mandatory (its Blakely, Georgia plant, the source of the outbreak, held a "superior" rating from its private auditor in the year before) and records produced at trial showed lots shipped after positive *Salmonella* tests on at least a dozen occasions in 2007–08, under falsified certificates of analysis; the convictions came under statutes that predate FSMA. What became the statute's centerpiece, the Preventive Controls for Human Food rule (21 CFR Part 117; compliance beginning September 2016, this paper's event), instead almost entirely codified the existing practices of the largest manufacturers: a written hazard analysis, environmental monitoring of the production environment (§117.165), hygienic zoning, and supplier verification.

I focus on the products exposed to *Salmonella*, the pathogen behind the outbreak that drove the reform: dry, low-water-activity, ready-to-eat foods, long treated as low-risk because they are dry, and brought under mandatory preventive controls only after the Act. My design leans on contrasts within product categories (those that share the shelf, have similar demand shocks, but have different regulatory burden from FSMA). Within a category (e.g., Peanut Butter, Nuts, Chocolate Bars), FSMA binds where either no *Salmonella* "kill step" exists in the plant, or a kill step (e.g., roasting) is followed by exposed handling (e.g., cooling and seasoning before packing) in which recontamination risk is present, and it does not bind when a *Salmonella* kill step exists at the end of the production line, or if the contamination monitoring is offloadable upstream (e.g., it is a responsibility of the farmer that supplies the processing plant). The Act's obligations recur with every production run. Environmental monitoring (§117.165) is swabbing and testing the processing environment wherever exposed product travels, on a schedule set by production; a positive triggers corrective action, re-sanitation, and retesting (§117.150). Hygienic zoning is kept shift by shift, in the movement of people and materials between the raw and the exposed sides of the plant. Every duty generates records to keep and review (§117.190). These are work hours, tests, and paperwork that increase with output, a marginal-cost incidence.

Regressive incidence was not an accident of drafting. The large branded manufacturers that lobbied for and supported the Act, had operated HACCP-style food-safety systems since the 1990s (i.e., hazard analysis, monitoring, documentation) in many cases as a precondition for exporting to the European Union and Japan. For them, the Preventive Controls rule largely codified existing practice: a hazard analysis they had already written, monitoring they already ran, at near-zero incremental cost. For facilities without prior programs, the same text meant new programs, new personnel, and a higher cost per unit of output. Compliance was staggered by size: the largest firms in September 2016, smaller tiers in 2017 and 2018, with very small (mostly organic) businesses largely exempt.

## 3.2 Data

The data are the Nielsen Retail Scanner data from the Kilts Center at the University of Chicago, 2012–2019. Participating stores report weekly revenue and units for every UPC sold (the Universal Product Code, the twelve-digit barcode that identifies a product). Stores span the grocery, drug, mass-merchandise, and convenience channels, all pooled here; the panel covers more than half of sales in U.S. grocery and drug stores and about a third of sales at mass merchandisers. Walmart and Costco do not participate. Supermarkets account for roughly 90% of revenue in the categories studied here.

Nielsen assigns every UPC to a product module, its finest product category (peanut butter, cold cereal, seasoning); the categories of this paper are modules, or partitions of a module when it pools distinct markets (Section 3.3). UPCs carry a brand, and brands are assigned to parent companies through an ownership crosswalk constructed from company ownership records as of 2016, so that later acquisitions do not reassign revenue across firms.[12] Store brands are excluded: Nielsen masks their UPCs to protect retailer identity, pooling all private-label products into a single pseudo-brand with no identifiable producer.

The sample keeps stores present in every year of the panel, 27,215 of 55,966, so that the set of stores is the same before and after the event. Retailers enter only if they report without interruption, 67 of 88, accounting for 99% of revenue. Table 1 traces the sample through these restrictions.

| | Stores | Retailers |
|---|---|---|
| Stores active in any year, 2012–2019 | 55,966 | 199 |
| Stores active in every year | 27,215 | 88 |
| Retailers reporting in all 32 quarters (99% of revenue) | 26,966 | 67 |

Table 1: Store sample construction. A store is active in a year if it records sales that year. Retailer counts are the retailers operating the stores in each row.

The unit of observation is the firm by county by quarter, built in three steps. Weekly revenue is observed by UPC and store; each UPC maps to its brand and, through the 2016 ownership crosswalk, to its parent firm; and revenue is summed over the firm's UPCs, over the stores in the county (taken from Nielsen's store location), and over the weeks of the quarter. A firm's county-quarter is kept at zero when the firm has sold in that county before and still sells elsewhere nationally; quarters in which a firm sells nowhere nationally are dropped. Zeros are between 5% and 33% of firm-county-quarter cells across categories, which the Poisson estimator of Section 3.4 accommodates. Baseline size is the firm's 2011 revenue in the category (Section 3.4). Table 2 reports, for each category, the number of firms, observations, the share of zero cells, the leading firm's revenue share, and category revenue.

[12] The crosswalk assigns some 2,700 brands to roughly 500 parent companies across the categories studied.

| Product | Firms | Obs. | Zero share | Leader share | Revenue 2016 ($M) |
|---|---|---|---|---|---|
| Nuts, oil-roasted | 24 | 518,392 | 0.14 | 0.45 | 142 |
| Nuts, dry-roasted | 34 | 604,924 | 0.21 | 0.64 | 273 |
| Nuts, raw | 36 | 380,619 | 0.24 | 0.39 | 136 |
| Dried fruit, true-dried | 20 | 424,678 | 0.29 | 0.42 | 96 |
| Dried fruit, fruit snacks | 18 | 516,413 | 0.20 | 0.43 | 225 |
| Peanut butter, creamy | 14 | 421,110 | 0.19 | 0.57 | 339 |
| Peanut butter, chunky | 12 | 284,799 | 0.21 | 0.47 | 77 |
| Chocolate, nut/PB | 35 | 1,196,003 | 0.23 | 0.38 | 868 |
| Chocolate, plain | 27 | 783,313 | 0.33 | 0.46 | 536 |
| Granola bars, w/ inclusions | 15 | 441,082 | 0.13 | 0.40 | 240 |
| Granola bars, plain | 10 | 339,588 | 0.18 | 0.70 | 120 |
| Granola cereal | 12 | 366,194 | 0.17 | 0.31 | 77 |
| Cereal | 25 | 615,285 | 0.05 | 0.37 | 2,042 |
| Seasoning | 35 | 967,067 | 0.15 | 0.56 | 385 |
| Crackers, flavored | 23 | 719,094 | 0.15 | 0.63 | 491 |

Table 2: Summary statistics by product. Firms and observations are those entering estimation (firms above the size floor, firm-county-quarter cells, 2016 excluded); zero share is the share of cells with zero revenue; leader share is the largest firm's share of 2016 revenue among sampled firms; revenue is 2016 revenue of sampled firms in the balanced-store panel.

### 3.3 Sample selection

(1) *Considered Categories*: the product is dry, low-water-activity, ready-to-eat, in the *Salmonella* low-moisture hazard class, with no terminal kill step at the end of the production line (frying, popping and cooking sterilize at the end of the line; such categories are excluded or serve as placebos when they have a paired treated category). (2) *Market coherence*: the product must be one competitive market. Several Nielsen modules are administrative assortments pooling non-substitutes with no clear partition that would make them a coherent substitutes market. (3) *Gradient estimability*: genuine size dispersion above the revenue floor; e.g., cases where the category consists of nine firms or fewer, or one brand holds 80% or more of all revenue, are excluded.

The rule removed: modules without enough firms to estimate a gradient (toaster pastries and most of the many cracker modules have fewer than six firms); sandwich cookies, where a single firm holds roughly 83% of revenue; and administrative residuals pooling non-substitutes (non-chocolate candy, which goes from hard candy to gummy bears, or the unclassified bulk of cookies, from chocolate chip to shortbread). Fifteen products pass the criteria; Table 2 lists them.

## 3.4 Estimation

*Specification.* The estimating equation (17) is fit by Poisson pseudo-maximum-likelihood on firm $\times$ county $\times$ time revenue (Santos Silva and Tenreyro, 2006):

$$r_{ict} \;=\; \exp\Big\{\,\beta\,\big(\ln \mathrm{size}_i^{2011} \times \mathrm{post}_t\big) \;+\; \delta_{ic} \;+\; \delta_{st}\,\Big\}\cdot \eta_{ict}, \qquad \mathbb{E}\big[\eta_{ict}\,|\,\cdot\,\big] = 1,$$

with firm-by-county and state-by-quarter fixed effects; $\beta$ is the change in the productivity–size gradient at the event (see Section 2.6). The estimator is consistent as long as the displayed moment condition holds, $\mathbb{E}\big[\eta_{ict}\,|\,\cdot\,\big] = 1$, with no other assumption on the distribution of revenue. The condition permits deviations that are heteroskedastic or serially correlated. Inference is by wild cluster bootstrap, clustered by firm.

*Baseline size.* Baseline size is 2011 firm revenue within category (the calendar year before the estimation panel) so the treatment variable cannot be mechanically linked to outcomes. A \$1M floor on 2011 size guarantees at least two years of compliance, so every sampled firm faced compliance by September 2017 at the latest. For firms between \$500k–\$1M, compliance starts in September 2018, and firms under \$500k fall under the "very small businesses" exemption. For the categories with very few firms, if a \$500k floor would bring the firm count above nine, then that floor is used instead.

*Time.* Calendar-year bins, reference year 2015. Compliance started in September 2016, making 2016 a straddle year (eight months pre, four post): all difference-in-differences and trend-break estimation have a donut design, therefore *dropping 2016* (pre = 2012–2015, post = 2017–2019). The event studies still display the 2016 coefficient.

*Anticipation.* The final rule text published in September 2015 and was widely covered in the trade press. The controls it mandates take time to put in place: supply-chain changes to accommodate monitoring, hygienic re-zoning of existing lines, verification programs negotiated with suppliers. Firms that started preparing to comply at publication would show responses before the September 2016 compliance date. This might be the reason we observe an effect starting to form in 2015 for most products.

*Products with linear pre-drift.* Products with flat pre-trends get the plain event study, summarized by the donut level difference-in-differences. Products with a linear pre-drift get the linear trend extrapolation of Freyaldenhoven et al. (2019): a linear trend is fit to the pre-period coefficients, extrapolated over the whole window, and subtracted from every coefficient, pre and post alike, so the adjusted path is net of the drift's counterfactual continuation. Two estimands summarize these products: the donut *trend break*, the change in the gradient per year at the event, and the trend-adjusted *post average*, the post-period level net of the extrapolated drift.

## 3.5 Results

The results are organized by contrasting products within categories: similar products that share shelf space and have similar demand shocks, but have different compliance burdens.

| Product | Role | $\hat{\beta}$ (level DiD) | s.e. | Obs. | Firms |
|---|---|---|---|---|---|
| Nuts, oil-roasted | treated | $+0.456^{***}$ | (0.092) | 518,392 | 24 |
| Nuts, dry-roasted | treated | $+0.238^{***}$ | (0.077) | 604,924 | 34 |
| Nuts, raw | placebo | $-0.075$ | (0.106) | 380,619 | 36 |
| Dried fruit, true-dried | treated | $+0.292^{***}$ | (0.096) | 424,678 | 20 |
| Dried fruit, fruit snacks | placebo | $-0.059$ | (0.065) | 516,413 | 18 |
| Peanut butter, creamy | treated | $+0.103^{***}$ | (0.039) | 421,110 | 14 |
| Peanut butter, chunky | treated | $+0.097^{**}$ | (0.050) | 284,799 | 12 |

Table 3: Flat-pre-trend products: donut level difference-in-differences (the post-2016 change in the productivity–size gradient, per log point of 2011 firm size, the productivity proxy). Poisson PML, firm×county and state×quarter fixed effects, wild cluster bootstrap by firm; 2016 dropped from estimation; \$1M size floor. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

| Product | Role | Trend break (/yr) | s.e. | Post avg. | Obs. | Firms |
|---|---|---|---|---|---|---|
| Chocolate, nut/PB | treated | $+0.014^{*}$ | (0.008) | $+0.039$ | 1,041,346 | 35 |
| Chocolate, plain | placebo | $-0.011$ | (0.026) | $-0.025$ | 681,387 | 27 |
| Granola bars, w/ inclusions$^{\dagger}$ | treated null | $+0.035$ | (0.041) | $+0.348^{*}$ | 382,778 | 15 |
| Granola bars, plain$^{\dagger}$ | placebo | $-0.046$ | (0.041) | $-0.073$ | 296,845 | 10 |
| Granola cereal | treated | $+0.112^{***}$ | (0.032) | $+0.256^{**}$ | 317,246 | 12 |
| Cereal | treated | $+0.036^{***}$ | (0.008) | $+0.064^{***}$ | 549,911 | 25 |
| Seasoning | treated | $+0.015^{***}$ | (0.006) | $+0.049^{***}$ | 840,400 | 35 |
| Crackers, flavored | treated | $+0.024^{**}$ | (0.012) | $+0.027$ | 621,482 | 23 |

Table 4: Linear-pre-drift products: donut trend break (the change in the productivity–size gradient per year at the event) and trend-adjusted post average, after the pre-trend removal of Freyaldenhoven et al. (2019). Poisson PML, firm×county and state×quarter fixed effects, wild cluster bootstrap by firm; 2016 dropped from estimation; \$1M size floor. $^{\dagger}$\$500K floor (the floor at which the product is estimable). $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

*Nuts:* The incidence in nuts will depend on whether the *Salmonella* control is the facility's own, non-offloadable §117.165 duty. A roaster runs a *Salmonella* kill step and then handles the exposed product *afterwards* (they have to let the nuts cool down before they salt/season them), so it must control recontamination in its own plant. A handler of *raw* nuts offloads the control to its supplier and runs supplier verification instead. This would predict a gradient change in roasted nuts and nothing in raw. Table 3 shows: dry-roasted +0.238

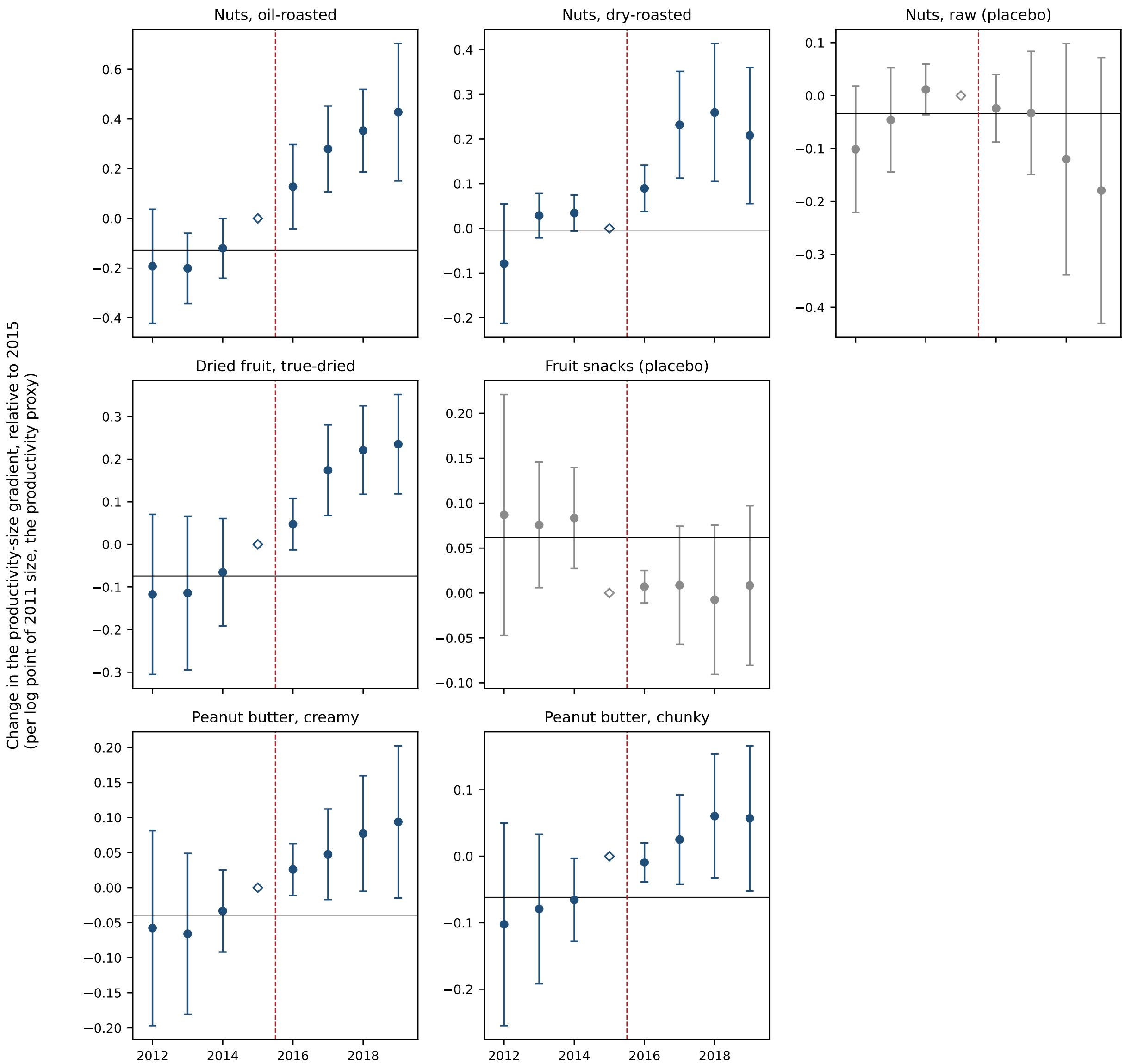


Figure 2: Event studies for the products of Table 3: the change in the productivity–size gradient by calendar year, with 95% confidence intervals (open diamond: reference year 2015; solid line: pre-period average, 2012–2015; dashed line: end of the pre period, with compliance from September 2016).

and oil-roasted +0.456 per log point of baseline size, against a raw-nut estimate of −0.075, indistinguishable from zero. In oil-roasted nuts, the estimated effect implies that two firms selling ten-to-one in 2011 ended the post period selling nearly thirty-to-one.

*Dried fruit:* True dried fruit needs to be monitored throughout its whole production. Fruit snacks, on the other hand, do not have a monitoring obligation, as they are cooked confections whose terminal cook sits at the end of the production line, with packaging im-

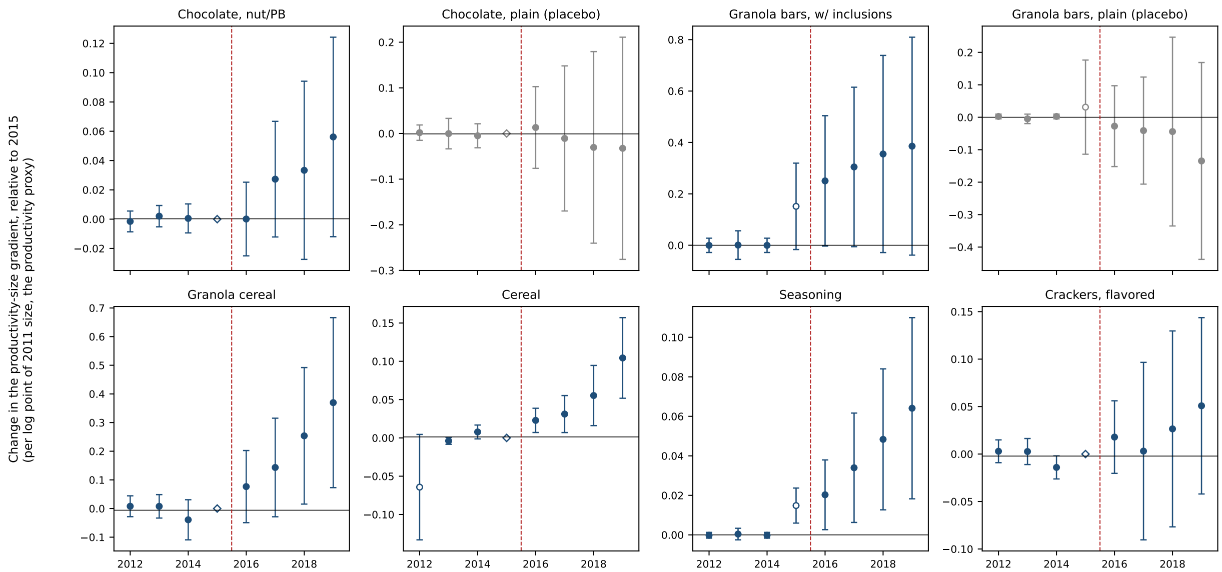


Figure 3: Event studies for the products of Table 4, shown after linear pre-trend removal; the trend fits exclude 2012 for cereal and 2015 for seasoning and granola bars (open markers), with the excluded years still plotted. Open diamond: reference year 2015; solid line: pre-period average, 2012–2015; dashed line: end of the pre period, with compliance from September 2016.

mediately after. The contrast delivers +0.292 against −0.059, a statistical zero.

*Peanut butter:* The statute's origin category; both the creamy and chunky versions are reported: +0.103 and +0.097, both significant. No paired placebo is available.

*Chocolate:* The chocolate category is the first paired category with a removed linear-pre-drift, it has both inclusion (treatment) and plain (placebo) varieties. An inclusion is an exposed, ready-to-eat ingredient added *after* the chocolate's own kill step. The manufacturer therefore carries the in-plant handling of the exposed inclusion. Plain chocolate's kill step sits upstream at cocoa roasting, and the process after it runs enclosed, with little exposed handling. Inclusion chocolate shows a (marginally) significant post-2016 gradient steepening (+0.014 per year, $p = 0.08$) and plain chocolate none (−0.011, $p = 0.68$; Table 4).

*A treated null:* Granola bars with nut, peanut-butter, fruit, or chocolate inclusions (the same inherited obligation as chocolates with inclusions) against a plain oats-and-honey placebo: the treated event study does ramp up after the event, directionally in line with the mechanism; but on the trend-break estimand it is a null (+0.035, $p = 0.39$; although the trend-adjusted post average is significant +0.348, $p = 0.06$).

*Other (linear-pre-trend):* The gradient becomes more positive year after year through the post period: cereal's grows by +0.036 per log point per year, granola cereal's by +0.112, seasoning's by +0.015, flavored crackers' by +0.024, all significant (Table 4).

## 3.6 The politics

The Grocery Manufacturers Association — whose members included General Mills and Kellogg (cereal), Mars, Hershey, and Nestlé (chocolate), J.M. Smucker (peanut butter), McCormick (seasoning), and Mondelēz and Kraft (crackers) — supported the bill throughout its passage, filing forty lobbying disclosures over its life, and celebrated the signing as "the most comprehensive reform of our nation's food safety laws in more than 70 years," one that would "help restore the public's faith in the safety and security of the food supply."[13] Opposition came from the other end of the size distribution: the very small producers said the burden would be insurmountable. By their own arithmetic, compliance would consume 4–6% of a small operation's gross sales, more than half its profits at the sector's typical margins.[14] The Organic Consumers Association's grassroots campaign delayed the Senate vote, and was resolved by the Tester amendment, which exempts "qualified facilities", those with average annual sales below $500,000, the majority sold directly to consumers or to restaurants and retailers in the same state or within 275 miles (21 U.S.C. §350g(l)).

Individual large firms also lobbied on the FSMA Act. Disclosures on the FSMA bill (LobbyView; Huneeus and Kim, 2024) show, alongside the GMA's forty filings: *Blue Diamond Growers* (the dominant almond processor) with fifty filings, in the category where the roasted-nut gradient is estimated; the *American Peanut Shellers Association* with forty-one, in peanut butter, the statute's origin category; *Mars*, *Nestlé*, and *Hershey* (44, 37, and 28 filings) in chocolate; *General Mills*, *Kellogg*, and *Post* in cereal, where the gradient also rises; and the *American Bakers Association* with fifty-six filings across the baked categories.

The regulation that passed would not have prevented the *Salmonella* outbreak that was used to justify it, and was backed by the GMA, which represents the largest firms in the industry, the ones whose practices were codified. The change in gradient estimates shows that the regulation was regressive. The mechanism proposed in this paper states that this is enough for the most productive firms to grow (gain revenue, profit, and market share), even under general equilibrium closure, as the next section shows.

[13] Pamela Bailey, GMA president and CEO, on the signing of FSMA; quoted in Helena Bottemiller, "Historic Food Safety Bill Signed into Law," *Food Safety News*, January 4, 2011. GMA's board chairman the same day: "I am proud of the food industry for its support of landmark food safety legislation."

[14] National Sustainable Agriculture Coalition, "Smaller Farms Likely to Face Higher Food Safety Compliance Costs" and companion FSMA cost analyses: compliance at 4–6% of gross sales for operations under $500,000, against a national average net farm income near 10% of sales, i.e., "more than half" of small-operation profits. The coalition filed thirty-five FSMA lobbying disclosures of its own.

# 4 The General Equilibrium Closure

Section 2 showcased the mechanism in partial equilibrium under homothetic single-aggregator preferences. This section brings it to general equilibrium by putting it in a nested CES setting. After a regressive incidence regulation hits a single industry, the most productive firms grow (gain revenue, profit, and market share) under the assumption that the outer-nest elasticity satisfies $\sigma_0 \leq 1$, standard in both the empirical estimates and the quantitative models (Atkeson and Burstein, 2008; Comin et al., 2021; Andreyeva et al., 2010).[15] Using this same setting, I measure the welfare cost of the estimated elasticity of incidence from the FSMA empirical application.

## 4.1 A closed economy

A representative household supplies $L$ units of labor inelastically, owns all firms, and has nested CES preferences over industries $h = 1, \ldots, H$: an outer nest with elasticity $\sigma_0$ and positive weights $\alpha_h$ over industry composites, each composite a CES aggregator with elasticity $\sigma_h > 1$ over its industry's varieties. Labor is the only factor and the wage $w = 1$ is the numeraire. The household spends its income $Y$, wages plus profits, and the outer nest allocates the share

$$S_h = \frac{\alpha_h P_h^{1-\sigma_0}}{\sum_k \alpha_k P_k^{1-\sigma_0}}, \qquad P_0 = \left(\sum\nolimits_h \alpha_h P_h^{1-\sigma_0}\right)^{\frac{1}{1-\sigma_0}}, \tag{18}$$

to industry $h$, where $P_h$ is the industry's CES price index and $P_0$ the economy's. A firm in industry $h$ with price $p$ earns revenue $r = (p/P_h)^{1-\sigma_h} S_h Y$.

The regulated industry $g$ is the CES industry of Section 2.4, with elasticity $\sigma_g$. Index its firms by productivity $\varphi$. A firm of productivity $\varphi$ uses $1/\varphi$ units of labor per unit of output, so its marginal cost is $\psi = w/\varphi \in [\psi_{\min}, \infty)$. The mass of firms is fixed and every firm charges the markup $\mu_g = \sigma_g/(\sigma_g - 1)$. The regulation is compliance labor: monitoring, zoning, and verification are hours, hired on the same labor market at the wage, so the labor requirement rises to $(1 + \varepsilon(\psi))/\varphi$, and marginal cost to $\psi^R = (1 + \varepsilon(\psi))\, w/\varphi = (1 + \varepsilon(\psi))\, \psi$. Incidence is regressive, with $\varepsilon$ continuous and strictly increasing in $\psi$, and $\varepsilon(\psi_{\min}) = \varepsilon_0 \geq 0$.

**Assumption 1** (Inelastic outer nest)**.** $\sigma_0 \leq 1$.

Households do not substitute easily between industries of the economy,[16] with the Cobb–Douglas nest ($\sigma_0 = 1$) being the workhorse assumption of the quantitative literature.

[15] The result survives $\sigma_0 > 1$ under the conditions of Appendix C: incidence vanishing for the most productive firms and $\sigma_g > \sigma_0$.

[16] Across broad sectors, Comin et al. (2021) estimate the price elasticity of their demand system at 0.26

## 4.2 The industry's expenditure grows

Only industry $g$'s price index moves as a result of the regulation. The index over the industry's firms is $P_g = \left(\int (\mu_g\,\psi)^{1-\sigma_g}\, dG(\psi)\right)^{1/(1-\sigma_g)}$, and the regulation scales each cost by its incidence factor, so

$$\left(\frac{P_g^R}{P_g}\right)^{1-\sigma_g} \;=\; \frac{\int_{\psi_{\min}}^{\infty} \psi^{1-\sigma_g}\,\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\, dG(\psi)}{\int_{\psi_{\min}}^{\infty} \psi'^{\,1-\sigma_g}\, dG(\psi')} \;=\; \int_{\psi_{\min}}^{\infty} s(\psi)\,\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\, dG(\psi), \quad (19)$$

where $s(\psi) = \psi^{1-\sigma_g}\big/\int \psi'^{\,1-\sigma_g}\, dG(\psi')$ is a firm's pre-regulation revenue relative to the industry's average, so that the weights $s(\psi)\, dG(\psi)$ integrate to one. By Jensen's inequality, $\Delta \ln P_g > \ln(1+\varepsilon_0)$, i.e., the index rises by more than the incidence of the most productive firms. Write $\nu \equiv (P_g^R/P_g)^{1-\sigma_0} \geq 1$, the factor by which industry $g$'s weight in the outer allocation, $\alpha_g P_g^{1-\sigma_0}$, changes.[17] Only that one term moves in the sum defining $P_0$, so dividing the post-regulation $(P_0^R)^{1-\sigma_0}$ by the pre-regulation one $P_0^{1-\sigma_0}$ turns each unaffected industry's summand into its pre-regulation share $S_h$, by (18):

$$D \;\equiv\; \left(\frac{P_0^R}{P_0}\right)^{1-\sigma_0} \;=\; \sum_{h\neq g} S_h \;+\; S_g\,\nu \;=\; 1 - S_g + S_g\,\nu \;\geq\; 1, \quad (20)$$

the factor by which the sum of all allocation weights, $P_0^{1-\sigma_0}$, changes; $D$ exceeds one strictly when $\sigma_0 < 1$. Substituting $P_g^R$ into (18), with $P_h^R = P_h$ for every $h \neq g$,

$$S_g^R \;=\; \frac{\alpha_g\,(P_g^R)^{1-\sigma_0}}{\sum_k \alpha_k\,(P_k^R)^{1-\sigma_0}} \;=\; \frac{\nu\,\alpha_g\,P_g^{1-\sigma_0}}{D\,P_0^{1-\sigma_0}} \;=\; \frac{S_g\,\nu}{D}, \qquad \Delta \ln S_g \;=\; \ln(\nu/D) \;\geq\; 0: \quad (21)$$

expenditure share increases for the regulated industry, since $\nu - D = \nu - 1 - S_g(\nu-1) = (1-S_g)(\nu-1) \geq 0$.

Income is wages plus profits. Under CES, the industry $h$'s margin (the share of revenue paid out as profit) is $1/\sigma_h$; write $m_g \equiv 1/\sigma_g$ and

$$m_{-g} \;\equiv\; \sum_{h\neq g} \frac{S_h}{1-S_g}\,\frac{1}{\sigma_h},$$

the expenditure-weighted average margin outside $g$. The regulation will not change $m_{-g}$

on U.S. household data and 0.57 on a panel of 39 countries; Herrendorf et al. (2013) place their benchmark elasticity at 0.9. The workhorse nested structure imposes a Cobb–Douglas outer nest across industries with CES within (Atkeson and Burstein, 2008; Hsieh and Klenow, 2009; Arkolakis et al., 2012). Within food, a review of 160 U.S. studies finds mean across-category price elasticities of 0.27–0.81 (Andreyeva et al., 2010).

[17] Industry $g$'s weight is $\alpha_g P_g^{1-\sigma_0}$, so $\alpha_g (P_g^R)^{1-\sigma_0}/\alpha_g P_g^{1-\sigma_0} = (P_g^R/P_g)^{1-\sigma_0} = \nu$.

since after industry $g$'s index moves, the $h \neq g$ industries' shares all scale by the same factor, so the weights $S_h/(1-S_g)$ remain constant. Then

$$Y \;=\; \frac{wL}{1-\kappa}, \qquad \kappa \;\equiv\; S_g\, m_g + (1-S_g)\, m_{-g}, \qquad \kappa^R - \kappa \;=\; \big(S_g^R - S_g\big)\big(m_g - m_{-g}\big)\,. \quad (22)$$

$\kappa$ is the economy's profit share of spending, the expenditure-weighted average of the industries' margins; wages constitute the remaining share of spending, so income is the wage bill scaled up by $1/(1-\kappa)$. Income $Y$ moves with the profit composition of spending. It rises if the regulated industry has the higher margin (i.e., $Y^R > Y$ if $m_g > m_{-g}$), because spending would be migrating toward the higher-margin industry, and falls in the opposite case. The results below hold regardless of what happens to income.

**Lemma 1** (Industry expenditure never falls)**.** *$B \equiv \Delta \ln(S_g Y) \geq 0$, with equality only at $\sigma_0 = 1$.*

Proof in Appendix A.

Even if income would decrease in equilibrium, the rise in the industry's share will more than compensate.

**Proposition 4** (Collective profitability)**.** *For $\sigma_0 < 1$, industry $g$'s aggregate revenue and profit rise strictly, and every other industry's fall strictly. For $\sigma_0 = 1$, they remain unchanged.*

Proof in Appendix A.

A cost-raising rule is collectively profitable for the regulated industry, and the transfer comes from the other industries' expenditure. The elasticity of incidence of Section 2.6 determines how unevenly the enlarged industry expenditure reshuffles.

## 4.3 Who wins

A firm with cost $\psi$ has revenue $r(\psi) = \big(\mu_g \psi / P_g\big)^{1-\sigma_g} S_g\, Y$ before the regulation and $r^R(\psi) = \big(\mu_g \psi^R / P_g^R\big)^{1-\sigma_g} S_g^R\, Y^R$ after. In logs,

$$\begin{aligned} \ln r(\psi) \;&=\; (1-\sigma_g)\big[\ln \mu_g + \ln \psi - \ln P_g\big] \;+\; \ln S_g \;+\; \ln Y\,, \\ \ln r^R(\psi) \;&=\; (1-\sigma_g)\big[\ln \mu_g + \ln \psi + \ln(1+\varepsilon(\psi)) - \ln P_g^R\big] \;+\; \ln S_g^R \;+\; \ln Y^R\,, \end{aligned}$$

differencing, the markup and the firm's own base marginal cost cancel out:

$$\Delta \ln r(\psi) \;=\; (\sigma_g - 1)\big[\Delta \ln P_g - \ln(1+\varepsilon(\psi))\big] \;+\; B\,, \quad (23)$$

with $B = \Delta \ln S_g + \Delta \ln Y$, the change in the industry's expenditure (Lemma 1). The first term is the partial-equilibrium revenue change of Section 2, where a firm gains when its own incidence lies below the rise in the (industry's) price index (13); $B$ is what the general equilibrium closure adds, common to every firm.

**Theorem 2** (Who wins in general equilibrium)**.** *$\Delta \ln r(\psi)$ is strictly decreasing in $\psi$, and firm $\psi$ gains revenue and profit if and only if*

$$\ln\bigl(1+\varepsilon(\psi)\bigr) \; < \; \Delta \ln P_g \; + \; \frac{B}{\sigma_g - 1}\,.$$

*The threshold strictly exceeds $\ln(1+\varepsilon_0)$, so the most productive firms always grow.*

Proof in Appendix A.

Because demand across industries is inelastic, the regulation pulls expenditure into the industry (Lemma 1), and that expenditure raises every firm's revenue in the same proportion, increasing the partial-equilibrium threshold (Prop. 1) by $B/(\sigma_g - 1)$. General equilibrium therefore raises every firm's revenue change by the same amount: winners gain more, losers lose less, and the winner set enlarges.

**Corollary 2** (Every firm can gain)**.** *Suppose $\sigma_0 < 1$ and the incidence schedule is bounded, $\varepsilon(\psi) \leq \varepsilon_{\max}$ for all $\psi$. If*

$$\ln(1+\varepsilon_{\max}) \; < \; \Delta \ln P_g + \frac{B}{\sigma_g - 1},$$

*then every firm in industry $g$ gains revenue and profit. At $\sigma_0 = 1$ or with unbounded $\varepsilon(\psi)$, the least productive firms always lose.*

The above inequality holds when the incidence is nearly proportional, i.e., a near-zero elasticity of incidence. No relative cost moves much, and the expenditure drawn into the industry makes every firm grow. Proof in Appendix A.

*Remark* 2 (Cobb–Douglas outer nest)*.* At $\sigma_0 = 1$ the outer nest fixes expenditure shares: $D = 1$, $B = 0$, and the regulation is zero-sum across the industry's firms, winners' gains equal to losers' losses. The winner condition collapses to the partial-equilibrium one (13). An inelastic nest turns the zero-sum reshuffle into a positive-sum one for the industry.

*Remark* 3 (Elastic outer nest)*.* For $\sigma_0 > 1$, expenditure leaves the industry as its price rises. The most productive firms would still gain under two conditions: their own incidence vanishes, $\varepsilon_0 = 0$, and varieties substitute more easily within the industry than across industries, $\sigma_g > \sigma_0$. Appendix C states the result; with heterogeneous margins it asks in addition that $\sigma_h \geq 2$ for every industry.

## 4.4 The welfare cost

Welfare is real income, $W = Y/P_0$. Since only industry $g$'s index moves,

$$\Delta \ln P_0 \;=\; \frac{\ln D}{1-\sigma_0} \;\geq\; S_g\,\Delta \ln P_g, \qquad \Delta \ln W \;=\; \Delta \ln Y \;-\; \Delta \ln P_0, \tag{24}$$

with $\Delta \ln Y = \ln\big[(1-\kappa)/(1-\kappa^R)\big]$ from (22) and $\Delta \ln P_0 = S_g\,\Delta \ln P_g$ at $\sigma_0 = 1$.[18]

**Proposition 5** (The regulation is a welfare loss)**.** *Under a common margin ($m_g = m_{-g}$), $\Delta \ln W = -\Delta \ln P_0 < 0$. With heterogeneous margins, the loss holds to first order in the index change $\Delta \ln P_g$:*

$$\Delta \ln W^{(1)} \;=\; -\,S_g\,\Delta \ln P_g\left[1-(1-\sigma_0)(1-S_g)\,\frac{m_g-m_{-g}}{1-\kappa}\right] \;<\; 0\,.$$

Proof in Appendix A.

The loss is the regulation's resource cost. Compliance labor displaces production labor, the industry's index rises, and consumers pay more for the same basket. To first order the loss is the category's expenditure share times its index rise; the bracketed term depends on the income response and remains strictly positive.

The formulas ask only for aggregate and own-industry objects, the internal structure of the other industries never enters.

## 4.5 Quantification

Let productivity be inverse Pareto, $\Pr(\varphi < x) = (x/\varphi_{\max})^\theta$ on $(0, \varphi_{\max}]$. Marginal cost $\psi = w/\varphi$ is then Pareto, $\Pr(\psi > y) = (\psi_{\min}/y)^\theta$ with $\psi_{\min} = w/\varphi_{\max}$. Under this parameterization, a proportional regulation changes the scale parameter and a regressive one the shape parameter (Appendix E). Assume the elasticity of incidence is constant, as in Section 2.6,

$$1+\varepsilon(\psi) \;=\; \left(\frac{\psi}{\psi_{\min}}\right)^{\beta} \;=\; \left(\frac{\varphi_{\max}}{\varphi}\right)^{\beta}, \tag{25}$$

with $\beta$ the incidence elasticity and that $\varepsilon_\omega \to \varepsilon_0 = 0$ as $\omega \to 0$. A "non-vanishing" incidence of the most productive ($\varepsilon_0 > 0$) would be a proportional increase on top of this regressive incidence, neutral for who wins by Proposition 2 and adding a common cost to welfare, so the loss computed below is a lower bound.

[18] $D = (1-S_g)\cdot 1 + S_g\,\nu$ is a weighted arithmetic mean of 1 and $\nu$, so by the arithmetic–geometric mean inequality $D \geq 1^{1-S_g}\,\nu^{S_g} = \nu^{S_g}$; taking logs and dividing by $1-\sigma_0 > 0$ gives $\Delta \ln P_0 \geq S_g\,\Delta \ln P_g$.

Post-regulation marginal cost is $\psi^R = (1+\varepsilon(\psi))\,\psi = \psi_{\min}\,(\psi/\psi_{\min})^{1+\beta}$, whose distribution is

$$\Pr\big(\psi^R > y\big) \;=\; \Pr\Big(\psi > \psi_{\min}\,(y/\psi_{\min})^{\frac{1}{1+\beta}}\Big) \;=\; \Big(\frac{\psi_{\min}}{y}\Big)^{\frac{\theta}{1+\beta}} : \tag{26}$$

the regulation is a shift of the shape parameter, $\theta \to \theta/(1+\beta)$, so the marginal-cost tail fattens (i.e., higher marginal-cost dispersion). The index response (19) is closed form under the parameterization. Using the incidence schedule (25) and the Pareto density $dG = \theta\,\psi_{\min}^{\theta}\,\psi^{-\theta-1}\,d\psi$, we get:

$$\Big(\frac{P_g^R}{P_g}\Big)^{1-\sigma_g} \;=\; \frac{\displaystyle\int_{\psi_{\min}}^{\infty} \psi^{1-\sigma_g}\,\big(\psi/\psi_{\min}\big)^{\beta(1-\sigma_g)}\,dG(\psi)}{\displaystyle\int_{\psi_{\min}}^{\infty} \psi^{1-\sigma_g}\,dG(\psi)} \;=\; \frac{\theta+\sigma_g-1}{\theta+(1+\beta)(\sigma_g-1)},$$

and taking logs,

$$\Delta \ln P_g \;=\; \frac{1}{\sigma_g-1}\,\ln\frac{\theta+(1+\beta)(\sigma_g-1)}{\theta+\sigma_g-1}, \tag{27}$$

which enters (20)–(24) directly; given $(\theta, \sigma_g, \sigma_0, S_g)$, the welfare cost is then identified by the estimated $\beta$.

The incidence elasticities $\beta$ are the product-specific estimates of Section 3 (the difference-in-differences estimator for the flat-pre-trend products and the trend-adjusted post average for the others). The Pareto shape $\theta$ comes from each product's 2011 size distribution; under the Pareto parameterization, the log distance of a firm's baseline revenue from the largest firm's, $\ln(r_{\max}/r_i)$, is exponential with rate $\theta/(\sigma_g-1)$.[19] I estimate the rate by the reciprocal of the average log distance over the firms, the tail-index estimator of Hill (1975). The demand elasticity $\sigma_g$ is the median between-firm elasticity of Hottman et al. (2016), 3.9, estimated on the same scanner data. The category's expenditure share $S_g$ is its 2016 revenue, re-scaled to the national market, over personal consumption expenditure.[20] The outer elasticity governs how spending on a product category responds to its own price index, so I set $\sigma_0 = 0.5$ based on the category-level price elasticities for food, 0.27 to 0.81 across the studies reviewed by Andreyeva et al. (2010).

Table 5 reports the results. A firm grows (gains revenue, profit, and market share) when

[19] The log of a Pareto variable is exponential, i.e., $\Pr\big(\ln(\psi/\psi_{\min}) > t\big) = e^{-\theta t}$. Log revenue distance is just $(\sigma_g - 1)$ times log cost distance, $\ln(r_{\max}/r_i) = (\sigma_g-1)\ln(\psi_i/\psi_{\min})$; this implies that the log revenue distance is exponential with rate $\theta/(\sigma_g-1)$; for an exponential variable, the mean is the reciprocal of the rate. Appendix F details the estimation.

[20] Sampled revenue is first divided by the balanced stores' share of category revenue across all Nielsen stores (0.83 to 0.88), and then by the panel's coverage of food-at-home sales, about 42%, from Nielsen's channel coverage weighted by USDA's outlet shares. The resulting shares are between 0.002% and 0.05% of expenditure.

| Product | $\beta$ | $\theta$ | $\Delta \ln P_g$ (%) | $B$ (%) | Firms gaining (%) | Welfare loss ($M/yr) |
|---|---|---|---|---|---|---|
| Nuts, oil-roasted | 0.456 | 0.93 | 10.2 | 5.1 | 22 | 41 |
| Nuts, dry-roasted | 0.238 | 0.87 | 5.8 | 2.9 | 22 | 44 |
| Dried fruit, true-dried | 0.292 | 0.82 | 7.1 | 3.5 | 21 | 19 |
| Peanut butter, creamy | 0.103 | 0.80 | 2.7 | 1.3 | 22 | 26 |
| Peanut butter, chunky | 0.097 | 0.86 | 2.5 | 1.2 | 23 | 5 |
| Chocolate, nut/PB | 0.039 | 0.58 | 1.1 | 0.5 | 18 | 27 |
| Granola bars, w/ inclusions | 0.348 | 0.96 | 8.0 | 4.0 | 23 | 54 |
| Granola cereal | 0.256 | 1.37 | 5.5 | 2.8 | 29 | 12 |
| Cereal | 0.064 | 0.93 | 1.6 | 0.8 | 24 | 95 |
| Seasoning | 0.049 | 0.66 | 1.4 | 0.7 | 19 | 15 |
| Crackers, flavored | 0.027 | 0.63 | 0.8 | 0.4 | 19 | 10 |

Table 5: Quantification by treated product, $\sigma_g = 3.9$, $\sigma_0 = 0.5$. $\theta$ is the fitted shape before the regulation; $\Delta \ln P_g$ the change in the category's price index from (27); $B$ the change in expenditure on the category (Lemma 1); firms gaining is the share of firms, by count, below the cutoff of Theorem 2; the welfare loss is $-\Delta \ln W$ times 2016 personal consumption expenditure, under a common margin.

its incidence lies below the cutoff of Theorem 2: $\ln(1 + \varepsilon(\psi)) < c \equiv \Delta \ln P_g + B/(\sigma_g - 1)$. Under (25) the condition becomes $\beta \ln(\psi/\psi_{\min}) < c$. The winners are therefore the firms with $\psi/\psi_{\min} < e^{c/\beta}$. Using the Pareto CDF, the share of winners is then $1 - e^{-\theta c/\beta}$. The welfare loss is $-\Delta \ln W$ times personal consumption expenditure; to first order it is the category's national spending times its price-index rise. In oil-roasted nuts, the estimated elasticity of 0.456 fattens the marginal-cost tail from $\theta = 0.93$ to 0.64, raises the category's price index by 10%, draws 5% more expenditure into the category, and leaves the most productive fifth of firms as winners; the welfare loss is $41 million a year. Across the eleven treated products the losses sum to $347 million a year, with cereal, the largest category, contributing over a quarter of it. The share of firms that gain sits between a fifth and a third everywhere.

# 5 The Literature in Cost-Function Space

## 5.1 A common structure

The models this paper builds on, and each paper I could find in the cost-increasing-regulation literature, differ in how firms compete, e.g., Cournot, a dominant firm facing a fringe, monopolistic competition. They differ in the type of goods, i.e., homogeneous versus differentiated.

They all share, though, one and the same description of the firm. Firm $i$ solves

$$\max_{q_i} \;\; p_i\, q_i - C_i(q_i), \qquad C_i(q_i) \;=\; c_i\, q_i + F, \tag{28}$$

a *constant* marginal cost $c_i$, possibly heterogeneous across firms, plus a fixed cost $F$, common across firms when present and often absent. The raising-rivals'-costs tradition works with heterogeneous $c_i$ and, typically, $F = 0$ (Williamson, 1968; Salop and Scheffman, 1983, 1987); Melitz-type models write $c_i = w/\varphi_i$ for a productivity draw $\varphi_i$ and use the common $F$ to generate exit (Melitz, 2003; Macedoni and Weinberger, 2026).

A regulation, in this language, is a deformation of the distribution of the $c_i$ or a shift of the common fixed cost $F$, and the analysis is a comparative static with respect to that change. The marginal-cost deformations in differentiated varieties are this paper's subject. Before organizing the literature, I want to first explore when is it legitimate to *posit* the cost function (28) directly (as this paper and every related paper do). The next subsection gives the exact condition. And Section 5.4 uses the type of deformation and the type of goods to organize the related work.

## 5.2 Isolation: when cost functions are the primitives

Let firm $i$ have a production technology $q_i = f_i(x_i)$ mapping an input vector $x_i$ into output, and let it face factor prices $r$.

**Definition 2** (Isolated cost-minimization). *Firm $i$'s cost-minimization problem,*

$$C_i(q_i) \;\equiv\; \min_{x_i} \left\{ r \cdot x_i \;:\; f_i(x_i) \geq q_i \right\},$$

*is* isolated *if firm $i$ takes the factor prices $r$ as given and $r$ is unaffected by the choices of any individual firm.*

Isolation is compatible with both partial and general equilibrium, i.e., $r$ can be determined in equilibrium via factor-market clearing. All it requires is that no single firm can influence the factor prices.

**Proposition 6** (Reduction to cost-function space). *Suppose cost-minimization is isolated for every firm. Then firm $i$'s problem is equivalent to a two-stage problem: first compute $C_i(q_i)$, a function of own output and the given factor prices alone; then choose a point on the demand curve to maximize $p_i q_i - C_i(q_i)$. The industry equilibrium is characterized entirely by the cost functions $\{C_i\}$ and demand; every primitive source of firm heterogeneity enters only through $\{C_i\}$.*

The cost-minimization stage requires no knowledge of rivals at all: since $r$ does not depend on any individual firm's choice, $C_i$ is a function of own output and the given factor prices alone. Rivals enter only at the second stage, and only the way they do in any oligopoly: through the residual demand curve firm $i$ faces (in monopolistic competition, through the aggregator each firm takes as given). Duality then guarantees that the first stage recovers all production-relevant information. The formal argument is in Appendix A.

*Remark* 4 (The related work is already here). Melitz-type models take the productivity distribution as given and work with $\psi = w/\varphi$, constant returns technology with a common input cost $w$, and marginal-cost heterogeneity determined by the productivity term $\varphi$; the raising-rivals'-costs models posit constant marginal costs directly. Under isolation the production function is redundant, so starting from the cost function is without loss.

## 5.3 Affine form

The conditions so far allow us to work in cost function space $\{C_i\}$ directly, but to obtain the affine form (28) we need two further restrictions: constant returns in the variable inputs, under which producing $q$ units costs $q$ times the minimized cost of one unit, so variable cost is linear in output; and a common fixed overhead $F$. Then

$$C_i(q_i) = c_i\, q_i + F, \tag{29}$$

the marginal cost can be written as $c_i = \frac{e(r)}{\varphi_i}$ where $e(r)$ is the unit cost of the input bundle, when technologies differ only by productivity, as in the classic Melitz model. For a regulation that raises input prices $r$ to have an asymmetric incidence, we need for the input bundle to be firm-specific, so that $c_i = \frac{e_i(r)}{\varphi_i}$. Incidence on firm $i$ would then be the firm's expenditure share on that factor. In Appendix J, I show two settings where regulations that increase factor prices lead to regressive incidence: (1) a uniform tax on labor in an industry where capital intensity rises with productivity, and (2) a minimum wage regulation in a setting where firm-specific wages rise with productivity (as in Helpman et al. (2010)).

Next, I organize the relevant literature.

## 5.4 Four ways to create winners

The question descends from classic capture theory (Stigler, 1971; Peltzman, 1976), where regulations were about keeping rivals out: the captured regulation controlled entry, kept supply artificially low, generating rents for the existing firms. The next generation explored incidence over production costs instead, where some firms gain from a regulation because it burdens their rivals more. To the best of my knowledge, every model in this second generation sits in one of the four cells of Table 6. The most populated cell is homogeneous goods with marginal-cost incidence, the *raising rivals' costs* tradition. The homogeneous goods with fixed-cost incidence cell is nearly empty. Differentiated goods with fixed-cost incidence belongs to the *selection* tradition. And the few papers in the differentiated goods with marginal-cost incidence cell find that marginal-cost regulations are neutral. This paper poses that the neutrality does not come from the regulation being marginal-cost, but because these papers study a proportional incidence in a homothetic environment, which is neutral (Prop. 2).

| | **Homogeneous good** | **Differentiated goods** |
|---|---|---|
| **Marginal cost** | raising rivals' costs (Salop and Scheffman, 1983; Fowlie et al., 2016) | *this paper*, against a neutrality backdrop (Macedoni and Weinberger, 2026; Anouliès, 2017) |
| **Fixed cost** | exclusion (Hviid and Olczak, 2016) | selection (Macedoni and Weinberger, 2026; Rebeyrol, 2023) |

Table 6: How a cost-raising regulation creates within-industry winners.

**Homogeneous good, marginal cost.** The oldest and fullest cell. Williamson (1968) models the Pennington case, a union wage that burdens labor-intensive small mines more than the large ones that negotiated it. Salop and Scheffman (1983, 1987) give the tradition its name and its canonical structure, a dominant firm that profits from raising the marginal cost of a price-taking fringe; Michaelis (1994) moves the same logic into Cournot, where the low-cost firm gains market share from a regulation that burdens every firm asymmetrically. There is a modern environmental literature that builds on this cell. Fowlie et al. (2016) price carbon into the marginal cost of cement producers in a dynamic Cournot industry, where cleaner firms gain output from dirtier rivals; Fowlie (2009) obtains the same reallocation in a static oligopoly under incomplete regulation. Baccianti and Schenker (2022) bring the incomplete-coverage setting to general equilibrium, where the spared firms gain through an increase in markups. Not every paper in this cell requires strategic interaction. In electricity markets, a carbon price re-ranks the merit order and the single price rises with the marginal unit's cost, so clean inframarginal generators collect the difference as Ricardian rents; Fabra and Reguant (2014) measure this, finding near-complete pass-through. In Muñoz-García and Akhundjanov (2016) the incidence goes the other way, and the least efficient firm is the one that is favored by the regulation. Grey (2018) adds a second stage with lobbying, a firm that adopts a clean technology and then lobbies for the emissions tax its rival will pay; in Cai and Li (2020) a sufficiently large clean firm lobbies to increase pollution taxes; Kennard (2020) documents the same behavior in the climate-regulation record. The empirical settings from this classic tradition include the 1833 Factory Act, OSHA, the Clean Air Act, and modern cap-and-trade (Marvel, 1977; Oster, 1982; Maloney and McCormick, 1982; Landes, 1980; Pashigian, 1984; Bartel and Thomas, 1985, 1987; Curtis and Lee, 2018).

**Homogeneous good, fixed cost.** A thin cell. Hviid and Olczak (2016) show that raising rivals' fixed costs can profitably force exit even when the strategist's own fixed cost rises as much or more, because the fixed cost bites the participation constraint rather than the price. Becker and Henderson (2000) empirically study air-quality regulation (a fixed cost) whose compliance costs fall on new plants while incumbents are grandfathered in.

**Differentiated goods, fixed cost.** The selection tradition. In Melitz (2003) a common fixed cost and a productivity distribution generate an exit cutoff; anything that raises the fixed cost pushes the cutoff up, the least productive firms exit, and survivors absorb their demand. Macedoni and Weinberger (2026) attach a regulation to this machinery and derive its politics, a fixed-cost product standard that large firms gain from and lobby for. In Rebeyrol (2023), a regulation that raises every firm's fixed cost by the same amount still benefits the survivors who inherit the exiters' demand. Abel-Koch (2013) derives the politics of that mechanism in trade, where the largest firms lobby such regulations in. Andersen (2018) measures the welfare cost of regulation in this environment and finds that the direct compliance cost understates the welfare loss.

**Differentiated goods, marginal cost.** The cell where this paper belongs is populated by three non-results. Macedoni and Weinberger (2026) prove that a regulation acting on marginal costs benefits no firm of any size, so that none lobbies for it; Anouliès (2017) finds a cap-and-trade scheme neutral for revenues and composition. Both statements rely on the assumption that the cost is increased by a proportional factor. The same neutrality holds, unremarked, in Forslid et al. (2018): after firms re-optimize abatement, their emission tax multiplies every marginal cost by a common factor, making the marginal-cost channel of the regulation neutral. Section 2 shows this neutrality has nothing to do with the regulation being marginal-cost; it is a property of proportional incidence. What the cell was missing is the type of deformation present in the papers of the raising-rivals'-costs tradition, regressive incidence over marginal costs. Demirer et al. (2026) find that the European Union's General Data Protection Regulation has regressive incidence, a marginal-cost wedge that falls with firm size. The theoretical contribution of this paper in one line: raising rivals' costs, embedded in differentiated goods and brought to general equilibrium.

# 6 Welfare in Homogeneous Goods

This last exercise asks what the raising-rivals'-costs tradition would report about the welfare consequences of FSMA. The model is the tradition's simplest. The good is homogeneous, a low-cost leader faces $n$ identical higher-cost followers, and the firms play Cournot against linear demand $p = a - bQ$. The regulation raises the followers' marginal cost $c_f$ and spares the leader's cost $c_L$. Welfare is total surplus, the sum of consumer and producer surplus. Salop and Scheffman (1987, Proposition 5) established that the sign of the welfare change is ambiguous, because the regulation also moves output from high-cost to low-cost firms. The sign is determined by the leader's market share, the threshold has a closed form, and the leader crosses it in four of the categories of Section 3. Appendix M repeats the analysis in the dominant-firm structure of Salop and Scheffman (1987). Appendix N calibrates a differentiated demand system that reproduces the homogeneous model's price, shares, and markups.

## 6.1 The leader's share measures the cost gap

In the Cournot equilibrium each firm's first-order condition equates its price–cost difference to $b$ times its own quantity, $p - c_i = b\, q_i$. Dividing the leader's condition by a follower's,

$$\frac{q_L}{q_f} \;=\; \frac{p - c_L}{p - c_f}\,,$$

so the leader's share is determined by its cost relative to the followers'. Summing the $n + 1$ first-order conditions gives $b\,Q = (n+1)\,p - c_L - n\,c_f$, and substituting demand, $b\,Q = a - p$, solves for the price:

$$p \;=\; \frac{a + c_L + n\,c_f}{n + 2}\,.$$

Each quantity is then its price–cost difference over $b$, and taking ratios,

$$\frac{q_L}{q_f} \;=\; \frac{(a - c_L) + n\,(c_f - c_L)}{(a - c_L) - 2\,(c_f - c_L)}\,, \qquad s_L \;=\; \frac{(a - c_L) + n\,(c_f - c_L)}{(n+1)(a - c_L) - n\,(c_f - c_L)}\,,$$

where $s_L$ is the leader's share of output and of revenue. Both expressions increase in the cost gap $c_f - c_L$ relative to $a - c_L$. Given the number of followers, the leader's share is a one-to-one function of that relative gap. This is what lets us determine the sign of the welfare change knowing only the leader's share and the number of firms.

## 6.2 Why surplus can rise

Total surplus is the area under the demand curve minus production costs. A small increase in the followers' cost changes it in two ways. The followers pay the higher cost on every unit they produce, the direct burden. But output reallocates from the least to the most efficient firm. A unit change in output from firm $i$ moves surplus by $p - c_i$, what consumers pay for the marginal unit net of what it costs to make. Together,

$$dTS \;=\; -\,n\,q_f\,dc_f \;+\; \sum_i (p - c_i)\,dq_i\,,$$

the single-industry form of the aggregation in Baqaee and Farhi (2020). The leader expands and the followers contract. A unit of output that moves from a follower to the leader is valued at the difference between the two price–cost differences, $(p - c_L) - (p - c_f) = c_f - c_L$. Every unit reallocated is registered as a resource saving equal to the cost gap. Against this "credit" stand the direct cost of the burden and the contraction of total output, both proportional to the followers' output. A large enough gap makes the credit on the reallocated units outweigh both, and the regulation raises total surplus.

## 6.3 The threshold

**Proposition 7** (When the homogeneous model reports a gain)**.** *The marginal regulation ($dc_f > 0$, leader spared) changes total surplus by*

$$\frac{dTS}{dc_f} \;=\; \frac{n}{n+2}\left[\, q_L - (n+4)\,q_f \,\right],$$

*so total surplus rises if and only if*

$$\frac{c_f - c_L}{a - c_L} \;>\; \frac{n+3}{3n+8} \qquad \textit{equivalently} \qquad s_L \;>\; s_L^*(n) \;\equiv\; \frac{n+4}{2(n+2)} \;\in\; \left(\tfrac{1}{2},\, \tfrac{5}{6}\right].$$

Proof in Appendix M. The threshold depends on the primitives only through $(s_L, n)$, declines as the number of firms grows, converging to one half.

Section 4 computed the welfare change from the regulation in the differentiated economy, finding a loss in every product. Seen through the lens of the homogeneous model, FSMA resulted in welfare gains in four of those product categories. At the firm counts of Table 2, the threshold goes from 0.53 to 0.58, and the leader sits above it in dry-roasted nuts (Wonderful, 0.64 against 0.53), flavored crackers (Mondelez, 0.63 against 0.54), seasoning (McCormick, 0.56 against 0.53), and creamy peanut butter (Smucker's Jif, 0.57, at its threshold). A researcher who took the homogeneous model to these categories would report that the regulation raised welfare in each of them.

# 7 Conclusion

How can firms benefit from a regulation that raises production costs, their own included? When the incidence of a regulation is asymmetric, every price rises, but the price of the least burdened firms rises the least. Every firm becomes more expensive, some become relatively cheaper, and demand flows toward them. They grow, gaining revenue, profit, and market share. The mechanism does not require barriers to entry or rivals being forced out, and it is distinct from the output reallocation of *raising rivals' costs*, which it amplifies. I focus on regulations whose incidence decreases with productivity, those with *regressive incidence*, the ones where the most productive firms already run the required systems, use less of the regulated input, or pay above the proposed floor. In a differentiated industry with homothetic single-aggregator demand, a firm gains if and only if the price aggregator rises proportionally more than its own marginal cost, and under regressive incidence the most productive firms grow while the least productive shrink. Specializing to CES makes log revenue linear in log productivity, surfacing the *productivity–size gradient*, and the change in the gradient at the regulation estimates the elasticity of incidence. Proportional incidence and incidence over fixed costs leave the gradient unchanged, progressive incidence flattens it, and only regressive incidence steepens it.

I find evidence of the mechanism in the Food Safety Modernization Act. In Nielsen scanner data around the September 2016 compliance date, the productivity–size gradient steepens in the product categories where the *Salmonella*-control obligation was the facility's own, and does not move in the closely related products whose terminal kill step or supplier verification allows them to avoid the obligation. The reallocations are economically large: in oil-roasted nuts, two firms selling ten-to-one before the regulation ended the post period selling nearly thirty-to-one. The largest firms in these categories, and their trade associations, supported and lobbied for the statute; the very smallest producers fought for an exemption. Brought to general equilibrium in a nested CES economy with an inelastic outer nest, the mechanism intensifies, since the industry's higher price index draws expenditure in from the other industries. With Pareto marginal costs, the estimated coefficients map into the distribution's shape parameter, and the estimates imply category price indices up to ten percent higher, only the top fifth to third of firms gaining, and a welfare loss of at least $347 million a year across the treated categories. The homogeneous-good model of the raising-rivals'-costs tradition would report welfare gains in four of those categories, because it treats the reallocation toward the lower-cost leader as an efficiency gain.

The results change how a cost-raising regulation should be evaluated. The relevant question is not only how much a regulation costs in aggregate, or whether it formally applies to every firm, but how its incidence varies across the affected industry. A regulation can be universal in law and sharply asymmetric in effect, and the incidence schedule is itself an object of political economy, to be measured alongside the regulation's direct benefits. Regulation reshapes an industry by changing relative marginal costs, which changes relative prices, which reallocates demand, raising costs and concentrating the market in the same move. It picks winners and losers.

The natural next step is studying how a regulation gets created. This paper takes regulations as given and asks who wins. The political economy side of the question, how firms would lobby over the shape of the incidence schedule, is reserved for future work.

# A Proofs

**Theorem 1 (Who wins).** *Under a deformation $\psi \to \psi^R$, firm $\omega$ attains strictly higher equilibrium revenue, profit, and market share if and only if its relative marginal cost falls,*

$$\frac{\psi_\omega^R}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})},$$

*strictly lower if the inequality is reversed, and unchanged under equality.*

*Proof of Theorem 1.* Section 2.1 established that firm $\omega$'s equilibrium profit, revenue, and market share are strictly decreasing in its relative marginal cost $\psi_\omega/P$: profit by the envelope condition (7), revenue $Y s_\omega(z_\omega^*)$ and share $s_\omega(z_\omega^*)$ because the optimal relative price rises with relative cost while the share function falls. Each outcome therefore improves if and only if the regulation lowers the ratio. Writing $\rho \equiv P(\boldsymbol{\mu}\boldsymbol{\psi}^R)/P(\boldsymbol{\mu}\boldsymbol{\psi})$ and rearranging,

$$\frac{\psi_\omega^R}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \quad \Longleftrightarrow \quad \frac{\psi_\omega^R}{\psi_\omega} < \frac{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)}{P(\boldsymbol{\mu}\boldsymbol{\psi})} = \rho\,,$$

and the reversed and equality cases are the same rearrangement with the inequality reversed or replaced by equality. In the equality case every firm's relative cost is unchanged, so every relative price, every share, and, industry expenditure given, every revenue $s_\omega Y$ and profit are unchanged. $\square$

**Lemma A.1** (Strict monotonicity of the aggregator)**.** *Let $p'_\omega \geq p_\omega$ for almost every $\omega$, with strict inequality on a set of positive measure. Then $P(\mathbf{p}') > P(\mathbf{p})$.*

*Proof.* Evaluate the adding-up integral (2) for the profile $\mathbf{p}'$ at the old value $P(\mathbf{p})$. Since each $s_\omega$ is strictly decreasing, the integrand weakly falls everywhere and strictly falls on a set of positive measure, so the integral is strictly below one. The integral is continuous and strictly increasing in the aggregator (again $s'_\omega < 0$), so restoring adding-up requires a strictly larger value: $P(\mathbf{p}') > P(\mathbf{p})$. $\square$

**Proposition A.1** (Exact characterization)**.** *Let $\rho \equiv P(\boldsymbol{\mu}\boldsymbol{\psi}^R)/P(\boldsymbol{\mu}\boldsymbol{\psi})$. Every firm in some neighborhood $(0, \delta)$ of the top gains if and only if $\psi_\omega^R/\psi_\omega < \rho$ for all $\omega$ in that neighborhood. In limit form:* $\limsup_{\omega \downarrow 0} \psi_\omega^R/\psi_\omega < \rho$ *is sufficient, and* $\limsup_{\omega\downarrow 0} \psi_\omega^R/\psi_\omega \leq \rho$ *is necessary.*

*Proof.* By Theorem 1, firm $\omega$ gains iff $\psi_\omega^R/\psi_\omega < \rho$. If the limsup is strictly below $\rho$, the inequality holds on a neighborhood (sufficiency); if the limsup exceeds $\rho$, there are firms arbitrarily close to the top of the productivity distribution with $\psi_\omega^R/\psi_\omega > \rho$, so no neighborhood of gainers exists (necessity of the weak inequality; equality is consistent with gains when the ratio approaches $\rho$ from below). $\square$

Corollary B.2's conditions (i)–(ii) are the interpretable special case: vanishing burden forces the limsup to one, and any rise in the aggregator puts $\rho$ above it. The exact condition shows what is actually needed — the burden along the productivity top tail must fall short of the economy-wide price response — and that "vanishing" can be relaxed to "sufficiently small relative to the aggregate rise."

**Proposition 1 (Regressive incidence).** *Take a cost-increasing regulation and write its incidence as $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega$. Suppose the incidence is regressive: $\varepsilon_\omega \le \varepsilon_{\omega'}$ whenever $\omega < \omega'$, with strict inequality for some pair. Then there exist $0 < \bar\omega \le \bar\omega' < 1$ such that every firm in $(0,\bar\omega)$ strictly gains revenue, profit, and market share, and every firm in $(\bar\omega', 1)$ strictly loses.*

*Proof of Proposition 1.* Write $1+\varepsilon_0 \equiv \lim_{\omega\downarrow 0}(1+\varepsilon_\omega)$ and $1+\varepsilon_1 \equiv \lim_{\omega\uparrow 1}(1+\varepsilon_\omega) \in (1+\varepsilon_0, \infty]$; both limits exist by monotonicity of the incidence, and $\varepsilon_1 > \varepsilon_0$ by the strict pair. Since $\psi^R_\omega = (1+\varepsilon_\omega)\psi_\omega \ge (1+\varepsilon_0)\psi_\omega$ almost everywhere, with strict inequality on a set of positive measure, monotonicity, homogeneity of degree one, and Lemma A.1 give

$$P(\boldsymbol{\mu}\boldsymbol{\psi}^R) \;>\; P\big((1+\varepsilon_0)\,\boldsymbol{\mu}\boldsymbol{\psi}\big) \;=\; (1+\varepsilon_0)\,P(\boldsymbol{\mu}\boldsymbol{\psi}),$$

so $\rho > 1+\varepsilon_0$. Meanwhile $\psi^R_\omega/\psi_\omega \to 1+\varepsilon_0$ as $\omega \downarrow 0$, so $\limsup_{\omega\downarrow 0} \psi^R_\omega/\psi_\omega = 1+\varepsilon_0 < \rho$, and Proposition A.1 gives a segment of winners at the top; because the incidence is monotone, the set $\{\omega : 1+\varepsilon_\omega < \rho\}$ is an interval adjacent to 0, which is $(0,\bar\omega)$.

For the losing segment, suppose first $\varepsilon_1 = \infty$; then $\psi^R_\omega/\psi_\omega = 1+\varepsilon_\omega > \rho$ for every $\omega$ close enough to one. If instead $\varepsilon_1 < \infty$, then $\psi^R_\omega \le (1+\varepsilon_1)\psi_\omega$ almost everywhere, with strict inequality on a set of positive measure (monotonicity and the strict pair again), so the same three properties give the mirrored bound $P(\boldsymbol{\mu}\boldsymbol{\psi}^R) < (1+\varepsilon_1)\,P(\boldsymbol{\mu}\boldsymbol{\psi})$, i.e., $\rho < 1+\varepsilon_1$; since $1+\varepsilon_\omega \to 1+\varepsilon_1$ as $\omega \uparrow 1$, again $\psi^R_\omega/\psi_\omega > \rho$ for every $\omega$ close enough to one. In either case Theorem 1 gives strict losses there, and monotonicity of the incidence makes $\{\omega : 1+\varepsilon_\omega > \rho\}$ an interval adjacent to 1, which is $(\bar\omega', 1)$. $\square$

**Corollary 1 (Exit amplifies).** *Suppose, in addition to the conditions of Proposition 1, that firms must pay a fixed operating cost $F > 0$ to remain active, and that a positive measure of firms exits under the regulation. Then the winning firms in $(0,\bar\omega)$ that survive gain strictly more than in the no-exit case.*

*Proof of Corollary 1.* Let $\Omega^R \subset (0,1)$, with $|\Omega^R| < 1$, be the post-regulation active set, and let $P^R(\cdot)$ denote the aggregator defined by adding-up over $\Omega^R$: $\int_{\Omega^R} s_\omega\big(\mu_\omega\psi^R_\omega/P^R\big)\,d\omega = 1$. Evaluated at $P^R = P(\boldsymbol{\mu}\boldsymbol{\psi}^R)$ (the no-exit aggregator), the integral falls short of one, since the integrand is unchanged but the domain has shrunk. Because each $s_\omega$ is strictly decreasing in its argument, restoring adding-up requires $P^R(\boldsymbol{\mu}\boldsymbol{\psi}^R) > P(\boldsymbol{\mu}\boldsymbol{\psi}^R)$. Hence for every surviving $\omega \in (0,\bar{\omega}) \cap \Omega^R$,

$$\frac{\psi_\omega^R}{P^R(\boldsymbol{\mu}\boldsymbol{\psi}^R)} \;<\; \frac{\psi_\omega^R}{P(\boldsymbol{\mu}\boldsymbol{\psi}^R)} \;<\; \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})},$$

the second inequality being the conclusion of whichever of Corollaries B.2 and B.1, or Proposition 1, applies (with $(0,\bar{\omega})$ read as that result's winning set). The argument uses only that the active set shrank; the exit technology — variable profit falling below the operating fixed cost $F$ for the most-burdened firms — plays no role beyond generating $\Omega^R$. Since variable profit is strictly decreasing in relative cost and $F$ is unchanged by the regulation, profit net of $F$ rises strictly more than in the no-exit case. □

**Proposition 2 (Proportional increases are neutral).** *Let $\psi_\omega^R = (1+\varepsilon)\,\psi_\omega$ for all $\omega$ and some $\varepsilon > 0$. Then every firm's relative marginal cost is unchanged, and so are every firm's revenue, profit, and market share.*

*Proof of Proposition 2.* With $\psi_\omega^R = (1+\varepsilon)\,\psi_\omega$ for every $\omega$, the scaled price profile is again an equilibrium: when every price and the aggregator scale by $1+\varepsilon$, every relative price and every relative cost $\psi_\omega/P$ are unchanged, so every firm's optimum is unchanged. By linear homogeneity of the aggregator,

$$\rho \;=\; \frac{P\big((1+\varepsilon)\,\boldsymbol{\mu}\boldsymbol{\psi}\big)}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \;=\; 1+\varepsilon \;=\; \frac{\psi_\omega^R}{\psi_\omega} \qquad \text{for every } \omega:$$

the equality case of Theorem 1, which delivers unchanged shares and, industry expenditure given, unchanged revenue and profit levels. □

**Proposition 3 (Progressive incidence).** *Take a cost-increasing regulation and write its incidence as $1+\varepsilon_\omega \equiv \psi_\omega^R/\psi_\omega$. Suppose the incidence is progressive: $\varepsilon_\omega \geq \varepsilon_{\omega'}$ whenever $\omega < \omega'$, with strict inequality for some pair. Then there exist $0 < \bar{\omega} \leq \bar{\omega}' < 1$ such that every firm in $(0,\bar{\omega})$ strictly loses revenue, profit, and market share, and every firm in $(\bar{\omega}',1)$ strictly gains.*

*Proof of Proposition 3.* Define $\rho(a) \equiv P\big(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1})\big)/P(\boldsymbol{\mu}\boldsymbol{\psi})$. Monotonicity of $P$ gives $\rho(a) > 1$; and since $\psi_\omega \geq \psi_0$ implies $\psi_\omega + a \leq (1 + a/\psi_0)\,\psi_\omega$, monotonicity and homogeneity give $\rho(a) \leq 1 + a/\psi_0$ as well. The relative burden $g(\omega) \equiv (\psi_\omega + a)/\psi_\omega = 1 + a/\psi_\omega$ is continuous and strictly decreasing in $\omega$, with $g(\omega) \to 1 + a/\psi_0$ as $\omega \downarrow 0$ and $g(\omega) \to 1$ as $\omega \uparrow 1$ (the support of $G$ is unbounded). Moreover both bounds on $\rho(a)$ are strict: $\psi_\omega + a < (1 + a/\psi_0)\,\psi_\omega$ strictly wherever $\psi_\omega > \psi_0$, so Lemma A.1 applied to the comparison profile gives $\rho(a) < 1 + a/\psi_0$, and the symmetric comparison gives $\rho(a) > 1$. The intermediate value theorem then delivers a unique $\varepsilon \in (0,1)$ with $g(\varepsilon) = \rho(a)$. For $\omega < \varepsilon$, $g(\omega) > \rho(a)$: the firm's own proportional cost increase exceeds the aggregator's, and the losing case of Theorem 1 gives the conclusions. □

**Lemma 1 (Industry expenditure never falls).** *$B \equiv \Delta \ln(S_g Y) \geq 0$, with equality only at $\sigma_0 = 1$.*

*Proof of Lemma 1.* Write $E_g \equiv S_g Y$ for expenditure on industry $g$, so $B = \Delta \ln S_g + \Delta \ln Y$. From (21), $S_g^R = S_g \nu / D$, and since $D - S_g \nu = 1 - S_g$, also $1 - S_g^R = (1 - S_g)/D$. Substituting into $\kappa^R = S_g^R m_g + (1 - S_g^R)\, m_{-g}$,

$$D\, \kappa^R \;=\; S_g\, \nu\, m_g + (1 - S_g)\, m_{-g}.$$

From (22), $\Delta \ln Y = \ln\big[(1-\kappa)/(1-\kappa^R)\big]$, so $B = \ln\big[\nu(1-\kappa)\big] - \ln\big[D(1-\kappa^R)\big]$, and the sign of $B$ is the sign of

$$\begin{aligned}
\nu(1-\kappa) - D(1-\kappa^R) &= \nu - D - \nu\big[S_g m_g + (1-S_g) m_{-g}\big] + S_g \nu\, m_g + (1-S_g)\, m_{-g} \\
&= (\nu - D) - (1 - S_g)(\nu - 1)\, m_{-g} \\
&= (1 - S_g)(\nu - 1)(1 - m_{-g}),
\end{aligned}$$

using $\nu - D = (1 - S_g)(\nu - 1)$ in the last step. Each factor is nonnegative, $1 - m_{-g} > 0$ because every $\sigma_h > 1$, and $\nu - 1 > 0$ exactly when $\sigma_0 < 1$. Hence $B \geq 0$, with equality only when $\nu = 1$, that is, at $\sigma_0 = 1$. □

**Proposition 4 (Collective profitability).** *For $\sigma_0 < 1$, industry $g$'s revenue and aggregate profit rise strictly, and every other industry's fall strictly. For $\sigma_0 = 1$, they remain unchanged.*

*Proof of Proposition 4.* Let $E_g$ and $E_{-g}$ denote post-regulation expenditure on industry $g$ and on all other industries. Two equilibrium conditions determine them. Wages exhaust revenue net of profit, and labor is supplied inelastically at $w$:

$$wL \;=\; (1 - m_g)\, E_g + (1 - m_{-g})\, E_{-g}.$$

The outer nest fixes the expenditure ratio, $E_g / E_{-g} = S_g^R/(1 - S_g^R) = S_g \nu/(1 - S_g)$. Substituting $E_{-g} = E_g\,(1-S_g)/(S_g\nu)$ into the first condition and solving,

$$E_g \;=\; \frac{wL}{(1-m_g) + \dfrac{1-S_g}{S_g\,\nu}\,(1-m_{-g})}, \qquad E_{-g} \;=\; \frac{wL}{(1-m_{-g}) + \dfrac{S_g\,\nu}{1-S_g}\,(1-m_g)}.$$

$E_g$ is strictly increasing in $\nu$ and $E_{-g}$ strictly decreasing; margins are constant, so profits $m_gE_g$ and $m_{-g}E_{-g}$ move with revenue. Every other industry $h \neq g$ has share $S_h^R = S_h/D$, so its revenue changes by $-\ln D + \Delta \ln Y = \Delta \ln E_{-g} < 0$, the same for all $h \neq g$. For $\sigma_0 < 1$, $\nu > 1$ and the inequalities are strict; for $\sigma_0 = 1$, $\nu = D = 1$ and nothing moves. □

**Theorem 2 (Who wins in general equilibrium).** *$\Delta \ln r(\psi)$ is strictly decreasing in $\psi$, and firm $\psi$ gains revenue and profit if and only if*

$$\ln\!\big(1+\varepsilon(\psi)\big) \;<\; \Delta \ln P_g \;+\; \frac{B}{\sigma_g - 1}\,.$$

*The threshold strictly exceeds $\ln(1+\varepsilon_0)$, so the most productive firms always gain.*

*Proof of Theorem 2. Monotonicity and the cutoff.* In (23), $B$ and $\Delta \ln P_g$ are common to all firms and $\ln(1+\varepsilon(\psi))$ is strictly increasing in $\psi$, so $\Delta \ln r(\psi)$ is strictly decreasing. Setting (23) to zero gives the cutoff

$$\ln\!\big(1+\varepsilon^*\big) \;=\; \Delta \ln P_g \;+\; \frac{B}{\sigma_g - 1}\,;$$

firms with $\varepsilon(\psi) < \varepsilon^*$ gain and firms above it lose, in revenue and, at the constant markup, in profit $r/\sigma_g$.

*The threshold exceeds the top incidence.* Because $\varepsilon$ is strictly increasing in $\psi$ and more than one cost level carries positive revenue, $(1+\varepsilon(\psi))^{1-\sigma_g} \leq (1+\varepsilon_0)^{1-\sigma_g}$ at every cost level, with strict inequality on a set of positive share. Averaging with the weights of the power mean (19),

$$\Big(\frac{P_g^R}{P_g}\Big)^{1-\sigma_g} \;=\; \int_{\psi_{\min}}^{\infty} s(\psi)\,\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\,dG(\psi) \;<\; (1+\varepsilon_0)^{1-\sigma_g}\,;$$

raising both sides to the negative power $1/(1-\sigma_g)$ reverses the inequality, $\Delta \ln P_g > \ln(1+\varepsilon_0)$. With $B \geq 0$ from Lemma 1, the threshold strictly exceeds $\ln(1+\varepsilon_0)$.

*The most productive always gain.* Evaluating (23) at $\psi_{\min}$,

$$\Delta \ln r(\psi_{\min}) \;=\; (\sigma_g - 1)\big[\Delta \ln P_g - \ln(1+\varepsilon_0)\big] + B \;>\; 0.$$

$\square$

**Corollary 2 (Every firm can gain).** *Suppose* $\sigma_0 < 1$ *and the incidence schedule is bounded,* $\varepsilon(\psi) \leq \varepsilon_{\max}$ *for all* $\psi$*. If* $\ln(1+\varepsilon_{\max}) < \Delta \ln P_g + B/(\sigma_g - 1)$*, then every firm in industry* $g$ *gains revenue and profit. At* $\sigma_0 = 1$ *or with unbounded* $\varepsilon(\psi)$*, the least productive firms always lose.*

*Proof of Corollary 2.* By Theorem 2, firm $\psi$ gains if and only if $\ln(1+\varepsilon(\psi)) < \Delta \ln P_g + B/(\sigma_g - 1)$; under the hypothesis the right side exceeds $\ln(1+\varepsilon_{\max}) \geq \ln(1+\varepsilon(\psi))$ for every $\psi$. At $\sigma_0 = 1$, $B = 0$ and the condition reads $\ln(1+\varepsilon(\psi)) < \Delta \ln P_g$; since $\Delta \ln P_g$ is a share-weighted power mean of $\ln(1+\varepsilon)$ over firms whose incidence varies, it lies strictly below the largest incidence, so the firms at the largest incidence lose. If $\varepsilon(\psi)$ is unbounded, $\ln(1+\varepsilon(\psi))$ exceeds the finite cutoff for all $\psi$ large enough, and those firms lose. $\square$

**Proposition 5 (The regulation is a welfare loss).** *Under a common margin (*$m_g = m_{-g}$*),* $\Delta \ln W = -\Delta \ln P_0 < 0$*. With heterogeneous margins, the loss holds to first order in the index change* $\Delta \ln P_g$*:*

$$\Delta \ln W^{(1)} \;=\; -\,S_g\,\Delta \ln P_g \left[1 - (1-\sigma_0)(1-S_g)\,\frac{m_g - m_{-g}}{1-\kappa}\right] \;<\; 0\,.$$

*Proof of Proposition 5.* From (20), $\Delta \ln P_0 = \ln D/(1-\sigma_0)$, and $W = Y/P_0$ gives $\Delta \ln W = \Delta \ln Y - \ln D/(1-\sigma_0)$. Under a common margin, $\kappa^R = \kappa$ by (22), so $\Delta \ln Y = 0$ and $\Delta \ln W = -\ln D/(1-\sigma_0)$, strictly negative for $\sigma_0 < 1$ because $D > 1$; at $\sigma_0 = 1$ the ratio tends to $-S_g\,\Delta \ln P_g < 0$.

With heterogeneous margins, expand to first order in $u \equiv \Delta \ln P_g$. From the definitions,

$$\nu \;=\; e^{(1-\sigma_0)\,u} \;\approx\; 1 + (1-\sigma_0)\,u, \qquad D \;=\; 1 - S_g + S_g\,\nu \;\approx\; 1 + S_g\,(1-\sigma_0)\,u,$$

so $\Delta \ln P_0 = \ln D/(1-\sigma_0) \approx S_g\,u$, and from (21),

$$S_g^R - S_g \;=\; \frac{S_g\,(\nu - D)}{D} \;\approx\; S_g\,(1-S_g)\,(1-\sigma_0)\,u\,.$$

By (22), $\kappa^R - \kappa = (S_g^R - S_g)(m_g - m_{-g})$, and

$$\Delta \ln Y \;=\; \ln \frac{1-\kappa}{1-\kappa^R} \;\approx\; \frac{\kappa^R - \kappa}{1-\kappa} \;=\; S_g\,(1-S_g)\,(1-\sigma_0)\,u\,\frac{m_g - m_{-g}}{1-\kappa}\,.$$

Collecting terms in $\Delta \ln W = \Delta \ln Y - \Delta \ln P_0$,

$$\Delta \ln W^{(1)} \;=\; -\,S_g\,u\left[1-(1-\sigma_0)(1-S_g)\,\frac{m_g-m_{-g}}{1-\kappa}\right].$$

The bracket is positive. If $m_g \le m_{-g}$ it exceeds one. Otherwise $(1-\sigma_0)(m_g-m_{-g}) \le m_g - m_{-g} \le 1-m_{-g}$, and $(1-S_g)(1-m_{-g}) \le 1-\kappa$ because the difference $1-\kappa-(1-S_g)(1-m_{-g}) = S_g(1-m_g)$ is positive; so the subtracted term is strictly below one. $\square$

**Proposition 6 (Reduction to cost-function space).** *Suppose cost-minimization is isolated for every firm. Then firm i's problem is equivalent to a two-stage problem: first compute $C_i(q_i)$, a function of own output and the given factor prices alone; then choose a point on the demand curve to maximize $p_i q_i - C_i(q_i)$. The industry equilibrium is characterized entirely by the cost functions $\{C_i\}$ and demand; every primitive source of firm heterogeneity enters only through $\{C_i\}$.*

*Proof of Proposition 6.* Under isolation, $r$ is invariant to any individual firm's choices, so the program $\min_{x_i}\{r\cdot x_i : f_i(x_i) \ge q_i\}$ is well defined given $q_i$ alone, and its value $C_i(q_i)$ is independent of $(q_{-i}, x_{-i})$. Firm $i$'s full problem $\max_{q_i,x_i}\{p_i(q_i)\,q_i - r\cdot x_i : f_i(x_i) \ge q_i\}$ therefore separates: for any candidate output, cost-minimization is optimal input choice, and substituting its value function reduces the problem to $\max_{q_i}\{p_i(q_i)\,q_i - C_i(q_i)\}$, where $p_i(q_i)$ collects the product-market side (residual demand, or the aggregator in monopolistic competition) and depends on rivals only through their outputs. By Shephard's lemma, $C_i$ recovers all production-relevant information about $f_i$ and $r$; hence any equilibrium of the primitive game corresponds to an equilibrium of the game in which firms are endowed with $\{C_i\}$, and conversely. $\square$

**Corollary B.1 (Kink deformations, stated in Appendix B).** *Suppose there is $\bar\omega \in (0,1)$ such that $\psi^R_\omega = \psi_\omega$ for all $\omega \in (0,\bar\omega)$, while $\psi^R_\omega \ge \psi_\omega$ elsewhere with strict inequality on a set of positive measure. Then every firm in $(0,\bar\omega)$ strictly gains.*

*Proof of Corollary B.1.* Equilibrium prices are $\mu_\omega\psi_\omega$, so the deformation raises prices weakly everywhere and strictly on a set of positive measure; Lemma A.1 gives $\rho > 1$. On $(0,\bar\omega)$ the cost ratio equals one, so Theorem 1 applies at every point of the segment. $\square$

**Corollary B.2 (Vanishing burden, stated in Appendix B).** *Suppose (i) the aggregator rises, $\rho > 1$; and (ii) $\lim_{\omega\downarrow 0}\psi^R_\omega/\psi_\omega = 1$. Then there exists $\bar\omega > 0$ such that every firm in $(0,\bar\omega)$ strictly gains.*

*Proof of Corollary B.2.* By condition (i), $\rho > 1$ is a constant. By condition (ii), $\psi^R_\omega/\psi_\omega \to 1$ as $\omega \downarrow 0$, so there exists $\bar{\omega} > 0$ with $\psi^R_\omega/\psi_\omega < \rho$ on $(0, \bar{\omega})$; Theorem 1 concludes. $\square$

# B Stronger Deformation Shapes: Kinks, Vanishing Burdens, and Exit

Two stronger variants of the top-sparing deformation. Each statement is a corollary of Theorem 1; proofs are in Appendix A. Corollary 1 applies under both variants as well.

**Corollary B.1** (Kink deformations: an untouched top)**.** *Suppose there is $\bar{\omega} \in (0, 1)$ such that $\psi^R_\omega = \psi_\omega$ for all $\omega \in (0, \bar{\omega})$, while $\psi^R_\omega \geq \psi_\omega$ elsewhere with strict inequality on a set of positive measure. Then* every *firm in $(0, \bar{\omega})$ strictly gains.*

The entire untouched segment grows. A wage floor under rent-sharing (see Appendix J) binds only for firms paying below it, leaving marginal costs above the productivity cutoff literally unchanged; a compliance rule that codifies what the most productive firms already do leaves their costs literally unchanged. Whenever the burden has a kink (zero up to a cutoff, positive beyond), every firm on the untouched side is a winner.

Alternatively, the burden need not be exactly zero anywhere; it is enough that it *vanishes* for the most productive. The class is most easily exhibited parametrically: take a cost distribution with a fixed, strictly positive minimum marginal cost (Pareto being the canonical case) and fatten its tail. Every firm's cost rises, but the increase vanishes as it approaches the minimum marginal cost, which is where the most productive firms sit: $\psi^R_\omega/\psi_\omega \to 1$ as $\omega \downarrow 0$, while the aggregator, facing higher costs everywhere, rises (Lemma A.1). I call deformations with these two features *shape-changing*: in the parametric case the deformation is implemented exactly by moving the shape parameter. The Pareto parametric family serves as an illustration (Appendix D works out CES–Pareto in closed form); stated formally:

**Corollary B.2** (Vanishing burden)**.** *Suppose (i) the aggregator rises, $\rho > 1$; and (ii) $\lim_{\omega\downarrow 0} \psi^R_\omega/\psi_\omega = 1$. Then there exists $\bar{\omega} > 0$ such that every firm in $(0, \bar{\omega})$ strictly gains.*

*Remark* B.1 (What the conditions do not require). The corollary does not require a parametric family for $G$, a stochastic-dominance ranking of $\psi^R$ against $\psi$, or a functional form for demand beyond the monotonicity already assumed. A cost increase everywhere is not required either: any cost-increasing regulation satisfies (i), by Lemma A.1, but (i) also admits deformations that *lower* some firms' costs. The two conditions are stated directly on observables of the deformation (what happens to the aggregate price environment, and what happens at the top of the productivity distribution), which makes them checkable without solving for the equilibrium. The exact necessary-and-sufficient characterization, showing precisely what can and cannot be relaxed, is Proposition A.1 in Appendix A. Appendix J derives the ordered burden from primitives in which a *uniform* regulation (a common tax on labor) generates it through factor intensities.

# C The Elastic Outer Nest

When $\sigma_0 > 1$, expenditure leaves industry $g$ as its price index rises, and Lemma 1 no longer holds. The winner result survives under two conditions: the incidence of the most productive firms vanishes, and varieties substitute more easily within the industry than across industries.

**Theorem C.1** (Winners under an elastic outer nest). *Let $\sigma_0 > 1$ and $\sigma_g > \sigma_0$. Suppose incidence is regressive with $\varepsilon(\psi_{\min}) = 0$ and $\varepsilon(\psi) > 0$ on a set of positive revenue share. Then, under a common margin across industries, the most productive firm gains revenue and profit, with*

$$\Delta \ln r(\psi_{\min}) \;\geq\; (\sigma_g - \sigma_0)\,\Delta \ln P_g \;>\; 0,$$

*since $\Delta \ln r(\psi)$ is strictly decreasing in $\psi$, the winners are a top segment of the productivity distribution. With heterogeneous margins the same conclusions hold if $\sigma_h \geq 2$ for every industry $h$.*

The gap in the elasticities $\sigma_g - \sigma_0$ multiplies the full index change; the outer nest contributes the expenditure outflow, which the gap outweighs, and the income composition term, which the bound on the elasticities keeps small. A positive incidence of the most productive firms is not admissible here: with expenditure escaping, the most productive firms' own cost increase can exceed the aggregator's, and the vanishing incidence is what rules it out.

Evaluating (23) at $\psi_{\min}$, where $\varepsilon_0 = 0$, and substituting $\Delta \ln S_g = (1 - \sigma_0)\,\Delta \ln P_g - \ln D$ from (21) into $B = \Delta \ln S_g + \Delta \ln Y$ gives the identity the proof rests on:

$$\Delta \ln r(\psi_{\min}) \;=\; (\sigma_g - \sigma_0)\,\Delta \ln P_g \;-\; \ln D \;+\; \Delta \ln Y\,. \tag{30}$$

**Lemma C.1** (Bounds under an elastic nest). *For $\sigma_0 > 1$:*

*(i)* $D \leq 1$*, so* $-\ln D \geq 0$*;*

*(ii)* $\left|S_g^R - S_g\right| \leq -\ln D$*;*

*(iii)* $|\Delta \ln Y| \leq \dfrac{\bar{\mu}}{\min_h \sigma_h}\big(-\ln D\big)$*, where $\bar{\mu} \equiv \max_h \sigma_h/(\sigma_h - 1)$ is the largest markup. In particular $|\Delta \ln Y| \leq -\ln D$ whenever $\sigma_h \geq 2$ for every industry $h$, and $\Delta \ln Y = 0$ under a common margin.*

*Proof.* (i) $\nu = e^{(1-\sigma_0)\Delta \ln P_g} \leq 1$ since $\sigma_0 > 1$ and $\Delta \ln P_g > 0$, so $D = 1 - S_g + S_g\nu \leq 1$. (ii) From (21), $S_g - S_g^R = S_g(1-S_g)(1-\nu)/D \leq S_g(1-\nu)$, since $D \geq 1 - S_g$; and $-\ln D = -\ln\bigl(1-S_g(1-\nu)\bigr) \geq S_g(1-\nu)$. (iii) By (22), $|\Delta \ln Y| = \bigl|\ln \frac{1-\kappa}{1-\kappa^R}\bigr| \leq \frac{|\kappa^R-\kappa|}{\min\{1-\kappa,\, 1-\kappa^R\}} \leq \bar{\mu}\,|\kappa^R-\kappa|$, since $1-\kappa$ is the expenditure-weighted mean of the $1-m_h$ and never falls below $1/\bar{\mu}$; and $|\kappa^R-\kappa| = |S_g^R - S_g|\,|m_g - m_{-g}| \leq |S_g^R - S_g|/\min_h \sigma_h$, since both margins lie below $1/\min_h \sigma_h$. Combining with (ii) gives the bound; when $\sigma_h \geq 2$ for all $h$, $\bar{\mu} \leq 2 \leq \min_h \sigma_h$ and the factor is at most one. □

*Proof of Theorem C.1. The index rises.* $\Delta \ln P_g > \ln(1+\varepsilon_0) = 0$ by the power-mean step in the proof of Theorem 2, since the incidence is positive on a set of positive share.

*Common margin.* $\kappa^R = \kappa$ by (22), so $\Delta \ln Y = 0$; with $-\ln D \geq 0$ from Lemma C.1(i), the identity (30) gives

$$\Delta \ln r(\psi_{\min}) \;=\; (\sigma_g - \sigma_0)\,\Delta \ln P_g \;-\; \ln D \;\geq\; (\sigma_g - \sigma_0)\,\Delta \ln P_g \;>\; 0\,.$$

*Heterogeneous margins.* With $\sigma_h \geq 2$ for every $h$, Lemma C.1(iii) gives $|\Delta \ln Y| \leq -\ln D$, so $-\ln D + \Delta \ln Y \geq 0$ in (30) and the same bound holds.

*Single crossing.* Profit is $r/\sigma_g$ and rises with revenue. In (23) the index and $B$ are common across firms and $\ln(1+\varepsilon(\psi))$ is strictly increasing, so $\Delta \ln r(\psi)$ is strictly decreasing; with a positive value at $\psi_{\min}$, the winners are the segment of the distribution below the single crossing. □

# D CES–Pareto Illustration

This appendix works out the parametric illustration of Section 2.2 in closed form. Let demand be CES with elasticity $\sigma > 1$, so the aggregator is

$$P(\boldsymbol{\mu\psi}) = \left[\int_0^1 \left(\tfrac{\sigma}{\sigma-1}\,\psi_\omega\right)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}},$$

and let marginal costs be Pareto, $\psi_\omega = \psi_0(1-\omega)^{-1/\theta}$ with $\theta > \sigma - 1$ (lower $\theta$: fatter right tail, as in Section 4.5). The integral inside the aggregator is a power integral,

$$\int_0^1 \psi_\omega^{1-\sigma}\, d\omega \;=\; \psi_0^{1-\sigma} \int_0^1 (1-\omega)^{\frac{\sigma-1}{\theta}}\, d\omega \;=\; \psi_0^{1-\sigma}\,\frac{\theta}{\theta+\sigma-1}\,,$$

so $P(\boldsymbol{\mu\psi}) = \mu\,\psi_0\left[\theta/(\theta+\sigma-1)\right]^{\frac{1}{1-\sigma}}$ and

$$\frac{\psi_\omega}{P(\boldsymbol{\mu\psi})} \;=\; \frac{(1-\omega)^{-1/\theta}}{\mu\left[\dfrac{\theta}{\theta+\sigma-1}\right]^{\frac{1}{1-\sigma}}}, \tag{31}$$

the numerator firm-specific, the denominator an industry-level statistic of the distribution's shape. The scale $\psi_0$ has cancelled between numerator and denominator (an instance of Proposition 2).

A shape-changing regulation is a marginal *decrease* in $\theta$: every firm's cost rises, dispersion widens, and the proportional increase vanishes at the left tail of costs, conditions (i)–(ii) of Corollary B.2, from Appendix B. Differentiating the logarithm of (31) with respect to $\theta$,

$$\frac{\partial}{\partial\theta}\ln\frac{\psi_\omega}{P} \;=\; \frac{\ln(1-\omega)}{\theta^2} \;+\; \frac{1}{\theta\,(\theta+\sigma-1)}\,,$$

the first term from the firm's own cost and the second from the aggregator, using $\frac{1}{1-\sigma}\frac{\partial}{\partial\theta}\ln\frac{\theta}{\theta+\sigma-1} = -\frac{1}{\theta(\theta+\sigma-1)}$. The derivative is positive (so that *lowering* $\theta$ lowers the firm's relative cost) if and only if $\ln(1-\omega) > -\theta/(\theta+\sigma-1)$, so firm $\omega$ benefits if and only if $\omega < \bar{\omega}$, with the closed-form cutoff

$$\bar{\omega} \;=\; 1-\exp\!\left(-\frac{\theta}{\theta+\sigma-1}\right) \;\in\; (0,1)\,. \tag{32}$$

Comparative statics of the winning set: $\bar{\omega}$ is *decreasing* in $\sigma$ and *increasing* in $\theta$; as $\sigma\to\infty$, $\bar{\omega}\to 0$, and as $\theta\to\infty$, $\bar{\omega}\to 1-e^{-1}\approx 0.63$. The intuition for the $\sigma$ result: with highly substitutable varieties, expenditure (and hence the aggregator) is dominated by the barely-burdened top of the productivity distribution, so the aggregate price response $\rho = P^R/P$ is weak and few firms clear it. Conversely, low substitutability makes the aggregator respond to the heavily-burdened right tail of costs, enlarging the winning set.

# E Deformations and Pareto Parameters

Assume marginal costs are Pareto, $\Pr(\psi > y) = (\psi_{\min}/y)^\theta$, with scale $\psi_{\min}$ and shape $\theta$, and that the elasticity of incidence is constant,

$$1+\varepsilon(\psi) \;=\; (1+\varepsilon_0)\left(\frac{\psi}{\psi_{\min}}\right)^{\beta}, \qquad \beta > -1,$$

where $\varepsilon_0$ is the incidence of the most productive firms and $\beta > -1$ keeps the deformation increasing in $\psi$ and the deformed costs Pareto. Substituting, $\psi^R = (1+\varepsilon_0)\,\psi_{\min}\,(\psi/\psi_{\min})^{1+\beta}$, and

$$\Pr\big(\psi^R > y\big) \;=\; \Big(\frac{(1+\varepsilon_0)\,\psi_{\min}}{y}\Big)^{\frac{\theta}{1+\beta}}\;:$$

Pareto with scale $(1+\varepsilon_0)\,\psi_{\min}$ and shape $\theta/(1+\beta)$. The parameter pairs are in one-to-one correspondence, $\varepsilon_0$ the scale ratio minus one and $\beta$ the shape ratio minus one, so each statement below holds in both directions.

**Proposition E.1** (Scale). *The incidence is proportional ($\beta = 0$) if and only if the deformation changes the scale parameter only.*

**Proposition E.2** (Shape). *The incidence is regressive ($\beta > 0$) or progressive ($\beta < 0$) if and only if the deformation changes the shape parameter, fattening the tail in the regressive case, $\theta/(1+\beta) < \theta$, and thinning it in the progressive case. The scale parameter changes by the factor $1+\varepsilon_0$.*

**Proposition E.3** (Pure shape). *Among cost-increasing deformations, the shape parameter changes alone if and only if the incidence is regressive and vanishes for the most productive firms.*

*Proof of Proposition E.1.* The deformed distribution has scale and shape

$$\psi^R_{\min} \;=\; (1+\varepsilon_0)\,\psi_{\min}, \qquad \theta^R \;=\; \frac{\theta}{1+\beta}\,,$$

and the map $(\varepsilon_0, \beta) \mapsto (\psi^R_{\min}, \theta^R)$ is one-to-one. If $\beta = 0$, then $\theta^R = \theta$ and $\psi^R_{\min} = (1+\varepsilon_0)\,\psi_{\min}$: the scale moves and nothing else. Conversely, if the deformation changes the scale parameter only, then $\theta^R = \theta$ forces $1+\beta = 1$, so $\beta = 0$ and the incidence is $1+\varepsilon(\psi) = 1+\varepsilon_0$ at every cost level: proportional. □

*Proof of Proposition E.2.* The elasticity of the incidence schedule is $d\ln(1+\varepsilon(\psi))/d\ln\psi = \beta$, so the incidence is regressive exactly when $\beta > 0$ (incidence rising with cost is incidence falling with productivity) and progressive exactly when $\beta < 0$. By the display above, $\beta \neq 0$ if and only if $\theta^R = \theta/(1+\beta) \neq \theta$: the shape parameter moves. In the regressive case $1+\beta > 1$, so $\theta^R < \theta$ and the tail fattens; in the progressive case $1+\beta \in (0,1)$, so $\theta^R > \theta$ and it thins. In either case the scale parameter changes by the factor $1+\varepsilon_0$, directly from the display. □

*Proof of Proposition E.3.* The shape parameter changes alone if and only if $\psi^R_{\min} = \psi_{\min}$, that is, $\varepsilon_0 = 0$. Suppose the deformation is cost-increasing with $\varepsilon_0 = 0$. Cost increase requires $1+\varepsilon(\psi) = (\psi/\psi_{\min})^{\beta} \geq 1$ at every $\psi \geq \psi_{\min}$, which forces $\beta \geq 0$; and $\beta = 0$ with $\varepsilon_0 = 0$ is the identity, excluded by Definition 1. So $\beta > 0$: the incidence is regressive and vanishes for the most productive firms. Conversely, regressive incidence with a vanishing top is $\beta > 0$ with $\varepsilon_0 = 0$, which fixes the scale and moves the shape alone. A cost-increasing *progressive* deformation, by contrast, attains its largest incidence at $\psi_{\min}$ (with $\beta < 0$, the factor $(\psi/\psi_{\min})^\beta$ is largest there), so cost increase forces $\varepsilon_0 > 0$ and the scale moves with the shape. □

# F Estimating the Pareto Shape

[TODO]

# G The Mechanism Beyond HSA

The results of Section 2 depend on two properties of HSA demand. First, the *equivalence*: maximizing over the price is the same as maximizing over the relative price, and the optimum depends on the firm's cost through the single ratio $\psi_\omega/P$. Second, the *scaling*: equilibrium profit is expenditure $Y$, which the firm and the regulation leave unmoved, times a decreasing function of that ratio, so an improved ratio is an absolute gain. This appendix asks how far Propositions 1 and 2 travel when these properties weaken: first across the other two homothetic classes of Matsuyama and Ushchev (2017), then outside homotheticity. Notation is local to the appendix: $\mathcal{I}(\mathbf{p})$ denotes a class's ideal price index (its unit expenditure function), and $\mathcal{B}(\mathbf{p})$, $\mathcal{C}(\mathbf{p})$ its secondary aggregators.

## G.1 The other homothetic classes

Matsuyama and Ushchev (2017) characterize three pairwise disjoint classes of homothetic demand systems whose only common member is CES. HSA is one. The second, *homothetic with direct implicit additivity* (HDIA), defines utility implicitly by $\int \phi_\omega(x_\omega/U)\, d\omega = 1$ with each $\phi_\omega$ increasing and concave; demand is

$$x_\omega \;=\; \frac{Y}{\mathcal{I}(\mathbf{p})}\, (\phi'_\omega)^{-1}\!\Big(\frac{p_\omega}{\mathcal{B}(\mathbf{p})}\Big),$$

with $\mathcal{B}$ defined by the adding-up condition $\int \phi_\omega\big((\phi'_\omega)^{-1}(p_\omega/\mathcal{B})\big)\, d\omega = 1$. The third, *homothetic with indirect implicit additivity* (HIIA), defines the ideal index itself by an adding-up condition, $\int \theta_\omega(p_\omega/\mathcal{I})\, d\omega = 1$, and demand is $x_\omega = Y\, \theta'_\omega(p_\omega/\mathcal{I})/\mathcal{C}(\mathbf{p})$ with $\mathcal{C} = \int p_\nu\, \theta'_\nu(p_\nu/\mathcal{I})\, d\nu$. In both classes an atomistic firm's profit factors as a scaling common to all firms times a function of one ratio:

$$\pi_\omega = \frac{Y\,\mathcal{B}}{\mathcal{I}}\Big(z_\omega - \frac{\psi_\omega}{\mathcal{B}}\Big)(\phi'_\omega)^{-1}(z_\omega) \quad \text{(HDIA)}, \qquad \pi_\omega = \frac{Y\,\mathcal{I}}{\mathcal{C}}\Big(z_\omega - \frac{\psi_\omega}{\mathcal{I}}\Big)\theta'_\omega(z_\omega) \quad \text{(HIIA)}, \tag{33}$$

where $z_\omega$ is the price relative to the class's own aggregator. The equivalence therefore extends: the firm's problem depends on its cost only through $\psi_\omega/\mathcal{B}$ (HDIA) or $\psi_\omega/\mathcal{I}$ (HIIA), and under HIIA the sufficient statistic is the cost relative to the ideal price index itself. What does not extend automatically is the scaling: $\mathcal{B}/\mathcal{I}$ and $\mathcal{I}/\mathcal{C}$ are ratios of aggregators that move independently; by Matsuyama and Ushchev (2017, Propositions 2 and 3), they are constant only under CES.

**Lemma G.1** (The aggregators inherit the three properties)**.** *$\mathcal{B}$ under HDIA and $\mathcal{I}$ under HIIA exist, are unique, are linearly homogeneous, and rise when prices rise on a set of positive measure and fall nowhere.*

*Proof.* Each is defined by an adding-up condition whose integrand moves monotonically in the price and in the aggregator, in opposite directions ($(\phi')^{-1}$ is decreasing under HDIA; $\theta$ is increasing under HIIA, so the signs flip together). The three properties then follow directly: the integral crosses its target once (existence, uniqueness), scaling all prices and the aggregator together leaves every ratio unchanged (homogeneity), and a positive-measure price increase pushes the integral off its target, which only the aggregator can restore (monotonicity). □

**Proposition G.1** (Propositions 1 and 2 beyond HSA)**.** *Let the regulation have regressive incidence as in Proposition 1. In each class: (i) the most productive firms' sufficient statistic strictly falls and the least productive firms' strictly rises, so every relative outcome (relative price, relative quantity, profit relative to any rival) improves for the former and deteriorates for the latter; (ii) if in addition the deformation raises the competition aggregator proportionally at least as much as the ideal price index, $\Delta \ln \mathcal{B} \geq \Delta \ln \mathcal{I}$ under HDIA and $\Delta \ln \mathcal{I} \geq \Delta \ln \mathcal{C}$ under HIIA, then the most productive firms gain in absolute profit; (iii) a proportional deformation, $\psi^R_\omega = (1+\varepsilon)\psi_\omega$, leaves every firm's output, share, and profit unchanged.*

*Proof.* (i) Lemma G.1 supplies exactly the properties the proof of Proposition 1 uses, so that proof applies with the class's aggregator in place of $P$: every cost rises at least by the most productive firms' factor and some strictly more, the aggregator rises strictly more than that factor, and the statistic falls at the top; the mirrored bound handles the least productive. (ii) By (33) and the envelope theorem, equilibrium profit is the scaling ($Y\mathcal{B}/\mathcal{I}$ or $Y\mathcal{I}/\mathcal{C}$) times a strictly decreasing function of the statistic. Under the stated condition the scaling weakly rises, and by (i) the decreasing function strictly rises at the top. (iii) Guess that every equilibrium price scales by $1+\varepsilon$. Every statistic $\psi_\omega/\mathcal{B}$ (or $\psi_\omega/\mathcal{I}$) is unchanged by homogeneity, so every optimal relative price is unchanged and the guess is a fixed point; both aggregators scale by $1+\varepsilon$, the scaling ratios are unchanged, and (33) returns every profit unchanged. □

*Remark* G.1 (When the level condition holds). The condition in (ii) holds with equality for proportional deformations, identically in the symmetric Kimball (1995) case (where $\mathcal{B}$ and $\mathcal{I}$ are proportional in symmetric equilibrium), and identically under CES. To first order, both sides are weighted averages of the equilibrium price responses: the ideal index weights them by expenditure shares, the competition aggregator by the weights of its own adding-up condition. The condition therefore asks that the aggregator governing competition load at least as heavily on the burdened firms as the cost-of-living index does.

## G.2 Outside the homothetic classes

*Symmetric translog.* The symmetric translog of Feenstra (2003) is a member of HSA with the geometric mean of prices as the aggregator, so Section 2 applies to it without modification. The general, asymmetric translog is not: budget shares depend on every price through pairwise coefficients that cannot be collapsed into a single ratio, and the equivalence fails at the first step.

*Almost ideal demand.* The AIDS system of Deaton and Muellbauer (1980) is non-homothetic by design: real income enters the budget shares, so no aggregator of prices alone summarizes a firm's position, and the framework does not apply.

*Additively separable preferences.* In the variable-elasticity class of Zhelobodko et al. (2012), demand is $x_\omega = (u')^{-1}(\lambda p_\omega)$ with $\lambda$ the marginal utility of income. Taking $\lambda$ as given, the firm's problem depends on its cost through $\lambda\psi_\omega$, and a regressive deformation moves the statistic in the paper's direction: $\lambda$ rises with the price profile while the most productive firms' costs barely move. But $\lambda$ depends on income as well as prices and is not homogeneous of degree minus one in prices unless preferences are homothetic, so a proportional deformation moves $\lambda\psi_\omega$ across firms: Proposition 2 fails, and proportional cost increases redistribute.

*Linear demand.* In Melitz and Ottaviano (2008), maximized profit is $(\bar{p}-\psi_\omega)^2/(8b)$, with $\bar{p}$ the demand intercept: the sufficient statistic is a cost *difference*, not a ratio. The taxonomy changes accordingly. A proportional deformation is not neutral, because the intercept is a preference parameter that does not scale with costs. And an additive deformation, which under Proposition 3 redistributes against the top, here subtracts a common constant from every firm's $\bar{p} - \psi_\omega$, a proportionally smaller loss for the most productive: the additive case flips sign relative to the homothetic taxonomy.

*Nested CES.* With different elasticities across nests, nested CES belongs to none of the three classes, and a firm's position depends on two statistics, its cost relative to the nest index and the nest index relative to the economy's. This is the system of Section 4, where the paper handles it directly.

### G.3 Summary

| Demand system | Equivalence | Prop. 1 (levels) | Prop. 2 | Sufficient statistic |
|---|---|---|---|---|
| CES | yes | yes | yes | $\psi/P$ |
| Symmetric translog | yes | yes | yes | $\psi/P$ |
| Kimball | yes | yes (symmetric) | yes | $\psi/\mathcal{B}$ |
| HDIA | yes | under Prop. G.1(ii) | yes | $\psi/\mathcal{B}$ |
| HIIA | yes | under Prop. G.1(ii) | yes | $\psi/\mathcal{I}$ |
| Nested CES | within nest | Section 4 | Section 4 | $\psi/P_g$ and $P_g/P_0$ |
| Additively separable | given $\lambda$ | directional | no | $\lambda\psi$ |
| Asymmetric translog | no | — | — | none |
| Linear demand | no | additive flips sign | no | $\bar{p} - \psi$ |
| AIDS | no | — | — | none |

Table 7: How far the mechanism extends. Equivalence: the firm's problem depends on its cost through a single statistic. Levels: regressive incidence raises the most productive firms' absolute profit. Proposition 2: proportional increases are neutral.

## H The Estimating Equation with Endogenous Expenditure

This appendix rebuilds the derivation of Section 2.6 without holding expenditure and the price of the input bundle fixed. Let each take its own value by period, $Y^p$ and $e^p$ for $p \in \{\text{pre}, \text{post}\}$. Write

$$A^p \;=\; \ln Y^p + (\sigma - 1)\ln\frac{P^p}{\mu\, e^p}$$

for the common terms of (10) in period $p$, with $P^p$ the period's price index. By (10) and (15),

$$\ln r_\omega^{\text{pre}} \;=\; \left(A^{\text{pre}} - a_0\right) \;+\; \ln r_\omega^0\,,$$

and, adding the regulation's incidence with constant elasticity, $(\sigma-1)\ln(1+\varepsilon_\omega) = (\sigma - 1)\,c - \tau\left(\ln r^0_\omega - a_0\right)$,

$$\ln r^{\text{post}}_\omega \;=\; \left(A^{\text{post}} - a_0 - (\sigma-1)\,c - \tau\,a_0\right) \;+\; (1+\tau)\,\ln r^0_\omega\,.$$

Every term other than $\ln r^0_\omega$ is common to all firms within its period. Combining the two periods with the indicator $\text{post}_t$ and grouping as in Section 2.6 gives

$$\ln r_{\omega t} \;=\; \beta\left(\ln r^0_\omega \times \text{post}_t\right) \;+\; \delta_\omega \;+\; \delta_t\,, \qquad \beta = \tau\,,$$

with $\delta_\omega = \ln r^0_\omega$ and $\delta_t = \left(A^{\text{pre}} - a_0\right) + \left(A^{\text{post}} - A^{\text{pre}} - (\sigma-1)\,c - \tau\,a_0\right)\text{post}_t$: the estimating equation (17), unchanged. Movements in expenditure, the input price, and the price index change only the period constants, which the time effects absorb; the coefficient on the interaction is still the elasticity of incidence.

# I The Gradient in Two Benchmark Models

Section 2.5 claims that proportional incidence and fixed-cost selection leave the productivity–size gradient unchanged. This appendix derives the invariance in two benchmark models: a competitive industry producing a homogeneous good, where the gradient is a constant of the technology, and the canonical selection model of differentiated varieties, where fixed costs act only through the exit cutoff.

## I.1 A competitive industry: Hopenhayn (1992)

A single competitive industry produces a homogeneous good. The output price $p$ and the wage $w$ are determined in equilibrium; individual firms take both as given. A firm's state is its productivity $\varphi > 0$, which evolves according to a Markov transition $\nu(\,\cdot \mid \varphi)$, independent across firms. Each period, a firm with productivity $\varphi$ chooses labor $\ell \geq 0$ to maximize static profit,

$$\pi(\varphi, p, w) \;=\; \max_{\ell \geq 0}\left\{\,p\,f(\varphi,\ell) - w\,\ell\,\right\} \;-\; c_f,$$

where $f$ is strictly increasing and concave in $\ell$ and $c_f > 0$ is a fixed operating cost paid each period; the first-order condition $p\,f_\ell(\varphi,\ell^*) = w$ pins down optimal labor. Markets clear through two primitives: the output price satisfies $p = D(Q)$ for total industry output $Q$, with $D$ strictly decreasing, and the wage satisfies $w = W(N)$ for total labor demand $N$, with $W$ weakly increasing. These two functions are the only objects through which the rest of the economy enters the industry. At the start of each period a firm decides whether to continue or exit at zero scrap value; the continuation value

$$V(\varphi) \;=\; \pi(\varphi, p, w) \;+\; \beta_d \int \max\{V(\varphi'), 0\}\, \nu(d\varphi' \mid \varphi),$$

with discount factor $\beta_d$, defines an exit threshold under standard monotonicity assumptions on $\nu$. Potential entrants pay a sunk cost, draw initial productivity from a fixed distribution, and enter while the expected value covers the cost.

Specialize the technology to $f(\varphi, \ell) = \varphi\, \ell^{\alpha}$ with $\alpha \in (0,1)$, so productivity scales output multiplicatively and the labor elasticity is common across firms. The first-order condition gives $\ell^* = (\alpha p \varphi / w)^{1/(1-\alpha)}$, and substituting into revenue $r = p\, \varphi\, (\ell^*)^{\alpha}$,

$$\ln r(\varphi, p, w) \;=\; K \;+\; \frac{1}{1-\alpha} \ln p \;+\; \frac{1}{1-\alpha} \ln \varphi \;-\; \frac{\alpha}{1-\alpha} \ln w, \qquad K = \tfrac{\alpha}{1-\alpha} \ln \alpha. \tag{34}$$

This is the architecture of (10): log revenue is linear in log productivity, the gradient $1/(1-\alpha)$ is a constant of the technology, and the price and the wage sit in the intercept, common to all firms. From the first-order condition the wage bill is a constant fraction of revenue, $w\,\ell^* = \alpha r$, so every firm has the same labor cost share regardless of productivity, and profit is $\pi = (1-\alpha) r - c_f$.

**Proposition I.1** (The gradient cannot move)**.** *Under the technology above, the proportional change in revenue from any shift $(p, w) \to (p', w')$ is identical across firms,*

$$\Delta \ln r \;=\; \frac{\Delta \ln p}{1-\alpha} \;-\; \frac{\alpha\, \Delta \ln w}{1-\alpha}\,,$$

*and the gradient of log revenue in log productivity is $1/(1-\alpha)$ before and after. No shift in demand $D$, in labor supply $W$, or in the fixed cost $c_f$, individually or jointly, moves the gradient.*

*Proof.* Differencing (34) at fixed $\varphi$, the productivity term cancels, so $\Delta \ln r$ does not depend on $\varphi$; the slope of (34) in $\ln \varphi$ is $1/(1-\alpha)$ at any $(p, w)$. The fixed cost enters neither revenue nor its difference. □

**Corollary I.1** (Survivors: everybody wins or everybody loses)**.** *Revenue rises for all surviving firms if and only if $p'/p > (w'/w)^{\alpha}$, and falls for all of them otherwise. Profit changes inherit the common sign, $\Delta\pi(\varphi) = (1-\alpha)\,\Delta r(\varphi)$, the fixed cost dropping out of the difference. There is no combination of shifts in $D$ and $W$ under which some surviving firms gain while others lose.*

*Proof.* Revenue rises iff $\Delta \ln r > 0$, which rearranges to $\ln(p'/p) > \alpha \ln(w'/w)$; the condition does not depend on $\varphi$, so it holds for all firms or for none. Profit: $\pi = (1-\alpha)r - c_f$, and $c_f$ is unchanged. $\square$

What does vary across firms is the magnitude. The exact revenue difference is $\Delta r(\varphi) = \alpha^{\alpha/(1-\alpha)}\, \varphi^{1/(1-\alpha)}\, \gamma$, where $\gamma$ collects the price and wage terms and is independent of $\varphi$, so $|\Delta r(\varphi)| \propto \varphi^{1/(1-\alpha)}$: more productive firms swing more in absolute terms, in the same direction as everyone else.

A regulation in this economy can shift demand, shift labor supply, pin the wage from below (a wage floor above the market-clearing level raises the exit threshold, lowers output, and raises the price), or raise the fixed cost. Whatever the combination, it reaches firms only through $(p, w)$ and the exit margin: survivors' log revenues move in parallel, the exit threshold moves, and the gradient does not. For the estimating equation of Section 2.6, every response in this economy loads on the time effects and the survivor set; $\beta = 0$ is a theorem here.

## I.2 The selection model: Melitz (2003)

The demand side is the CES system of Section 2.4: a representative consumer with elasticity $\sigma > 1$ over a continuum of varieties, aggregate expenditure $Y$, price index $P$, and revenue $r(\omega) = Y\,(p(\omega)/P)^{1-\sigma}$ for a variety priced at $p(\omega)$.

Each firm draws a productivity $\varphi$ from a distribution $G$ on $(0, \infty)$. Production requires a fixed cost $F > 0$ in units of labor and $q/\varphi$ units of labor for $q$ units of output; with the wage as numeraire, marginal cost is $1/\varphi$. Every firm charges the CES markup, $p(\varphi) = \frac{\sigma}{\sigma-1}\,\frac{1}{\varphi}$, and revenue is a power function of productivity,

$$r(\varphi) \;=\; r_1\, \varphi^{\sigma-1}, \qquad r_1 \;\equiv\; Y\Big(\frac{\sigma}{\sigma-1}\Big)^{1-\sigma} P^{\sigma-1}, \tag{35}$$

with $r_1$ common to all firms. Operating profit is $\pi(\varphi) = r(\varphi)/\sigma - F$. A firm produces if and only if operating profit is non-negative, which defines the cutoff $\varphi^*$ by $r_1(\varphi^*)^{\sigma-1}/\sigma = F$; entrants pay a sunk cost $F_e$ to draw from $G$, and free entry makes expected profit net of the entry cost zero, $[1 - G(\varphi^*)]\,\bar{\pi}(\varphi^*) = F_e$. The equilibrium is the cutoff, the mass of active firms $M$, and the index $P$, determined by the zero-cutoff-profit condition, free entry, the price index over survivors, and labor-market clearing. Under a Pareto distribution, $G(\varphi) = 1 - (\varphi_{\min}/\varphi)^k$ with $k > \sigma - 1$, the objects have closed forms:

$$\bar{\pi} \;=\; \frac{F}{k-(\sigma-1)}, \qquad \varphi^* \;=\; \varphi_{\min}\Big(\frac{k-(\sigma-1)}{k}\,\frac{F_e}{F}\Big)^{-1/k}, \qquad M \;=\; \frac{L}{F\,\sigma k/(k-(\sigma-1))}\,.$$

**Proposition I.2** (Fixed costs move the cutoff, not the gradient)**.** *Raise the fixed cost $F$. The cutoff $\varphi^*$ rises and the least productive survivors exit. Among the firms that remain,* $\ln r = \ln r_1 + (\sigma-1)\ln\varphi$*: the common intercept $r_1$ adjusts through $P$ and $M$, and the gradient is $\sigma - 1$ before and after. Relative revenues are invariant,*

$$\frac{r(\varphi_1)}{r(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1} \qquad \textit{for any two survivors,}$$

*whatever happens to $P$, $M$, $Y$, or $f$. No surviving firm's relative position moves; the winners of a fixed-cost regulation are made on the extensive margin only.*

*Proof.* Under the Pareto parameterization the cutoff is $\varphi^* = \varphi_{\min}\big(\frac{k-(\sigma-1)}{k}\,F_e/F\big)^{-1/k}$, strictly increasing in $F$. The ratio claim needs no distributional assumption: by (35), $r(\varphi_1)/r(\varphi_2) = (\varphi_1/\varphi_2)^{\sigma-1}$, free of $r_1$, and the slope of $\ln r$ in $\ln\varphi$ is $\sigma - 1$ at any value of the aggregates. □

The exit margin is exactly the amplification channel of Corollary 1: exiters shrink the set over which the index aggregates, the index rises, and survivors absorb the exiters' demand in proportion to their shares. There is no reallocation among survivors through relative prices: relative prices are pinned by relative productivities and the common markup, so reallocation among survivors comes only through the index, common to all. A regulation that moves the gradient must therefore move marginal costs asymmetrically across firms, the deformation of Section 2.2. For the estimating equation, as in the competitive benchmark, a fixed-cost regulation loads on the time effects and the survivor set; $\beta = 0$ again.

# J Microfoundations of Regressive Incidence

The theorems of Section 2 take the deformation of marginal costs as the primitive. This appendix derives two types of deformations: regressive incidence and a kink, from production-side primitives, in each case from a labor-cost regulation: a uniform tax on labor for the regressive incidence, and a wage floor under rent-sharing for the kink. Nothing in either construction is specific to labor; any regulated input priced by a common factor price works identically.

## J.1 Heterogeneous factor intensity: a regressive incidence

Let firm $\varphi$ produce with the constant-returns technology $q = \varphi\, K^{\alpha(\varphi)} L^{1-\alpha(\varphi)}$, with the capital share $\alpha(\varphi)$ *increasing* in productivity, i.e., larger, more productive firms are more capital-intensive. Cost-minimization (isolated, per Proposition 6) gives the Cobb–Douglas unit cost

$$\psi(\varphi) \;=\; \frac{\chi\big(\alpha(\varphi)\big)\, r_K^{\alpha(\varphi)}\, w^{1-\alpha(\varphi)}}{\varphi}, \qquad \chi(\alpha) \;=\; \alpha^{-\alpha}\,(1-\alpha)^{-(1-\alpha)}, \qquad \frac{\partial \ln \psi(\varphi)}{\partial \ln w} \;=\; 1-\alpha(\varphi),$$

the log derivative being the firm's labor cost share. A *uniform* labor tax $t$, a proportional increase in the price of labor, $w \to (1+t)\,w$, therefore multiplies the unit cost by $(1+t)^{1-\alpha(\varphi)}$: the incidence is

$$1+\varepsilon(\varphi) \;=\; (1+t)^{1-\alpha(\varphi)},$$

strictly decreasing in $\varphi$ because $\alpha(\cdot)$ is increasing, so we get regressive incidence from a uniform regulation, with $\varepsilon(\varphi) \approx \big(1-\alpha(\varphi)\big)\,t$ for small $t$. With $\alpha(\varphi) \uparrow \bar{\alpha} \le 1$, the top-tail burden is $\varepsilon_0 = (1+t)^{1-\bar{\alpha}} - 1 \ge 0$, and Proposition 1 applies whenever $\alpha(\cdot)$ is not constant. Each firm is constant returns and is isolated, so the setting belongs to the class of models of Section 5.

### J.2 Rent-sharing wages: a kink

Alternatively, let all firms share the technology $q = \varphi L$ but pay productivity-dependent wages $w(\varphi) = w_0\,\varphi^{\gamma}$, $\gamma \in (0,1)$, the rent-sharing schedule derived from bargaining in Helpman et al. (2010), under which more productive firms pay more. Marginal cost is $\psi(\varphi) = w(\varphi)/\varphi = w_0\,\varphi^{\gamma-1}$. A wage floor $\bar{w}$ binds if and only if $w(\varphi) < \bar{w}$, i.e. below the productivity cutoff $\bar{\varphi} = (\bar{w}/w_0)^{1/\gamma}$:

$$\psi^{R}(\varphi) \;=\; \begin{cases} \bar{w}/\varphi & \varphi < \bar{\varphi} \quad \text{(bound by the floor)} \\ \psi(\varphi) & \varphi \ge \bar{\varphi} \quad \text{(literally unchanged),} \end{cases}$$

a continuous, two-piece deformation whose burden is zero above the cutoff and strictly positive below, which is the kink of Corollary B.1. Every firm above $\bar{\varphi}$ strictly gains. Because each atomistic firm takes its wage schedule as given, the construction is isolated (Definition 2) and stays in cost-function space.

## K The Minimum Wage as a Shape Change: Evidence

[TODO]

# L Cross-Industry Substitution: Nested CES in Partial Equilibrium

Does the mechanism survive demand leaking to *other* industries? This appendix embeds the treated industry in a nested-CES economy: consumers allocate expenditure across industries $g$ through an outer elasticity $\sigma_0$ and across firms within each industry through inner elasticities $\sigma_g > 1$. Aggregate expenditure $Y$ is held fixed; what the nest adds relative to Section 2 is the cross-industry substitution margin $S_g$.

Firm $j$'s profit in industry $g$ is $\pi_j^g = \frac{YS_g}{\sigma_g}\big(p_j/P_g\big)^{1-\sigma_g}$, with the industry's expenditure share $S_g = \alpha_g P_g^{1-\sigma_0} \big/ \sum_h \alpha_h P_h^{1-\sigma_0}$: when the regulation raises the industry price index $P_g$, expenditure now escapes to other industries.

**Proposition L.1** (Cross-industry substitution does not overturn the result)**.** *Consider a regressive incidence regulation with vanishing burden in industry $g$. The most productive firms' absolute profit rises if and only if*

$$\sigma_g - 1 \;>\; (\sigma_0 - 1)\,(1 - S_g),$$

*for which $\sigma_g > \sigma_0$ (i.e., goods within an industry are closer substitutes than goods across industries, the standard nesting assumption) is sufficient. The single-industry shock is the worst case: an economy-wide regressive incidence regulation with vanishing burden raises other industries' price indices too, which feeds expenditure back toward $g$ and relaxes the condition a fortiori.*

*Proof.* Take logs of profit, $\ln \pi_j^g = \ln(Y/\sigma_g) + \ln S_g + (1-\sigma_g)\big(\ln p_j - \ln P_g\big)$, and differentiate. At the top of the distribution the burden vanishes, $d\ln\psi_j = 0$, and at the constant markup $d\ln p_j = d\ln\psi_j = 0$, so the within-industry term contributes $(\sigma_g - 1)\,d\ln P_g$. For the share, differentiate $\ln S_g = \ln\alpha_g + (1-\sigma_0)\ln P_g - \ln\sum_h \alpha_h P_h^{1-\sigma_0}$ at the worst case $d\ln P_h = 0$ for $h \neq g$:

$$d\ln S_g \;=\; (1-\sigma_0)\,d\ln P_g \;-\; S_g\,(1-\sigma_0)\,d\ln P_g \;=\; (1-\sigma_0)\,(1-S_g)\,d\ln P_g\,.$$

Adding the two pieces,

$$d\ln\pi_j^g \;=\; \big[(\sigma_g - 1) + (1-\sigma_0)(1-S_g)\big]\,d\ln P_g\,,$$

and the bracket is positive if and only if $\sigma_g - 1 > (\sigma_0 - 1)(1 - S_g)$. For sufficiency of $\sigma_g > \sigma_0$: $\sigma_g - 1 > \sigma_0 - 1 \geq (\sigma_0 - 1)(1 - S_g)$. An economy-wide regulation adds terms $d\ln P_h > 0$ for

$h \neq g$, which enter $d \ln S_g$ with positive sign and relax the condition. □

# M Welfare in Homogeneous Goods: Derivations

## M.1 Cournot: proof of Proposition 7

**Proposition 7 (When the homogeneous model reports a gain).** *The marginal regulation ($dc_f > 0$, leader spared) changes total surplus by $dTS/dc_f = \frac{n}{n+2}\big[q_L - (n+4)\,q_f\big]$, so total surplus rises if and only if $(c_f - c_L)/(a - c_L) > (n+3)/(3n+8)$, equivalently $s_L > s_L^*(n) \equiv (n+4)/\big(2(n+2)\big) \in (\frac{1}{2}, \frac{5}{6}]$.*

Demand is $p = a - bQ$; the leader produces at marginal cost $c_L$ and the $n$ rivals at $c_f > c_L$; all $n+1$ firms choose quantities simultaneously. Each firm's first-order condition is $p - c_i = b\,q_i$, and the equilibrium is

$$q_L = \frac{(a-c_L) + n\,(c_f - c_L)}{(n+2)\,b}\,, \qquad q_f = \frac{(a-c_L) - 2\,(c_f - c_L)}{(n+2)\,b}\,, \qquad p = \frac{a + c_L + n\,c_f}{n+2}\,.$$

Differentiating total surplus $TS = \int_0^Q P(x)\,dx - c_L q_L - n\,c_f q_f$ with respect to $c_f$, using $P(Q) = p$ at the margin, gives the decomposition of Section 6.2,

$$\frac{dTS}{dc_f} = \sum_i (p - c_i)\,\frac{dq_i}{dc_f} - n\,q_f\,,$$

and the equilibrium quantities respond by

$$\frac{dq_L}{dc_f} = \frac{n}{(n+2)\,b}\,, \qquad \frac{dq_f}{dc_f} = -\frac{2}{(n+2)\,b}\,.$$

Substituting these and the first-order conditions $p - c_L = b\,q_L$, $p - c_f = b\,q_f$,

$$\frac{dTS}{dc_f} = b\,q_L\,\frac{n}{(n+2)\,b} - n\,b\,q_f\,\frac{2}{(n+2)\,b} - n\,q_f = \frac{n}{n+2}\,\big[\,q_L - (n+4)\,q_f\,\big]\,.$$

Substituting the equilibrium quantities into the bracket,

$$q_L - (n+4)\,q_f = \frac{-\,(n+3)\,(a-c_L) \;+\; (3n+8)\,(c_f - c_L)}{(n+2)\,b}\,,$$

positive exactly when $(c_f - c_L)/(a - c_L) > (n+3)/(3n+8)$, the first form of the proposition. For the second form, total output is $Q = q_L + nq_f = \big[(n+1)(a-c_L) - n\,(c_f - c_L)\big]/\big((n+2)\,b\big)$, so the leader's share is

$$s_L \;=\; \frac{(a-c_L)+n\,(c_f-c_L)}{(n+1)(a-c_L)-n\,(c_f-c_L)}\,,$$

increasing in $(c_f-c_L)/(a-c_L)$ since the numerator rises and the denominator falls. Writing $g \equiv (c_f-c_L)/(a-c_L)$, the share is $s_L = (1+ng)/\big((n+1)-ng\big)$; at the threshold gap $g^* = (n+3)/(3n+8)$ the numerator is $(n+2)(n+4)/(3n+8)$ and the denominator $2(n+2)^2/(3n+8)$, so

$$s_L^*(n) \;=\; \frac{(n+2)(n+4)}{2\,(n+2)^2} \;=\; \frac{n+4}{2(n+2)}\,,$$

declining from 5/6 at $n=1$ toward 1/2 as $n\to\infty$. Since the share is monotone in the gap, the two forms of the condition are equivalent. □

## M.2 The dominant firm

In the dominant-firm structure of Salop and Scheffman (1987) the leader chooses its quantity first and the $n$ rivals play Cournot among themselves given that quantity. With $A = a - c_L$ and $\Delta = c_f - c_L$, the equilibrium is $q_L = (A+n\Delta)/(2b)$ and $q_f = \big(A-(n+2)\Delta\big)/\big(2b\,(n+1)\big)$, with first-order conditions $p-c_f = b\,q_f$ for the rivals and $p-c_L = b\,q_L/(n+1)$ for the leader, whose margin carries the factor $n+1$ because it internalizes the rivals' response. The quantities respond by $dq_L/dc_f = n/(2b)$ and $dq_f/dc_f = -(n+2)/\big(2b\,(n+1)\big)$, and the same differentiation gives

$$\frac{dTS}{dc_f} \;=\; \frac{b\,q_L}{n+1}\cdot\frac{n}{2b} \;-\; n\,b\,q_f\cdot\frac{n+2}{2b\,(n+1)} \;-\; n\,q_f \;=\; \frac{n}{2(n+1)}\left[\,q_L-(3n+4)\,q_f\,\right],$$

so total surplus rises if and only if

$$\frac{c_f-c_L}{a-c_L} \;>\; \frac{2n+3}{4n^2+11n+8} \qquad \text{equivalently} \qquad s_L \;>\; \frac{3n+4}{4(n+1)} \;\in\; \left(\tfrac{3}{4},\,\tfrac{7}{8}\right].$$

Because the leader's margin carries the factor $n+1$, a given share corresponds to a smaller cost gap than in Cournot, and the share threshold is higher.

# N Matching the Homogeneous Model

Section 6 fits the homogeneous model to a category's leader share and firm count. This appendix shows that the same moments admit the differentiated model: a nested demand system can be calibrated to reproduce a homogeneous-good equilibrium, its price, its shares, and its markups, exactly, with every parameter pinned by the observables. The construction takes any equilibrium in which the industry sells at one price $p^*$, a leader with marginal cost $c_L$ holds share $s_L$, and $n$ followers with marginal cost $c_f \in (c_L, p^*)$ hold the rest; the Cournot equilibrium of Section 6.1 is one instance, and the Stackelberg variant below is a second, fully worked one.

## N.1 A worked homogeneous target: Stackelberg

Let the leader commit to its quantity first and let the $n$ followers play Cournot among themselves given that commitment, against the linear demand of Section 6. Write $A \equiv a - c_L$ and $\Delta \equiv c_f - c_L > 0$. Each follower maximizes $(p - c_f)\, q_i$ given the leader's quantity and the other followers'; at the symmetric follower equilibrium,

$$q_f(q_L) \;=\; \frac{A - \Delta - b\, q_L}{b\,(n+1)}\,.$$

Substituting into demand, the price the leader faces is $p(q_L) = \big(a + n\, c_f - b\, q_L\big)/(n+1)$, and maximizing $(p(q_L) - c_L)\, q_L$ gives

$$q_L \;=\; \frac{A + n\Delta}{2b}\,, \qquad p^* \;=\; \frac{a + (n+1)\, c_L + n\, c_f}{2(n+1)}\,, \qquad s_L \;=\; \frac{(n+1)(A + n\Delta)}{(2n+1)A - n\Delta}\,.$$

At $n = 0$ this is the monopoly solution; at $\Delta = 0$ it is the symmetric Stackelberg outcome, the leader holding $(n+1)/(2n+1) > 1/2$ of the market. The Lerner indices are $(p^* - c_L)/p^*$ for the leader and $(p^* - c_f)/p^*$ for each follower, differing by $\Delta/p^*$.

## N.2 The differentiated side

Nest the leader's variety and the followers' varieties in a two-tier CES: an upper tier with elasticity $\sigma_U$ and weights $\alpha_L, \alpha_F$ over the leader's good and the follower composite, the composite itself a CES aggregator with elasticity $\sigma_F$ over the $n$ follower varieties. Firms are atomistic in the sense of Section 2.1: each takes the relevant aggregator as given. The follower composite's price index at symmetric follower prices $p_f$ is $P_F = n^{1/(1-\sigma_F)}\, p_f$, and the pricing conditions are the CES markups tier by tier,

$$p_L \;=\; \frac{\sigma_U}{\sigma_U - 1}\, c_L\,, \qquad p_f \;=\; \frac{\sigma_F}{\sigma_F - 1}\, c_f\,,$$

with Lerner indices $1/\sigma_U$ and $1/\sigma_F$.

## N.3 The calibration is exactly determined

Matching the two Lerner indices to the homogeneous equilibrium's pins both elasticities:

$$\sigma_U = \frac{p^*}{p^* - c_L}, \qquad \sigma_F = \frac{p^*}{p^* - c_f},$$

and with them both prices equal $p^*$. Since $c_L < c_f < p^*$, both elasticities exceed one and $\sigma_U < \sigma_F$: the leader sits in the less elastic tier, its higher markup absorbing its lower cost, which is what lets two different costs sell at one price. With one price, expenditure shares are quantity shares, and the upper-tier share equation pins the weight ratio,

$$\frac{e_L}{1 - e_L} = \left(\frac{\alpha_L}{\alpha_F}\right)^{\sigma_U} n^{-\frac{1-\sigma_U}{1-\sigma_F}} = \frac{s_L}{1 - s_L} \quad \Longrightarrow \quad \frac{\alpha_L}{\alpha_F} = \left[\frac{s_L}{1 - s_L}\, n^{\frac{1-\sigma_U}{1-\sigma_F}}\right]^{1/\sigma_U},$$

each follower receiving $(1 - s_L)/n$ within its nest at equal costs. Industry expenditure $Y = p^* Q^*$ matches the scale. Every parameter is a function of the observables $(p^*, c_L, c_f, s_L, n, Q^*)$; nothing is free. The calibrated system reproduces the homogeneous equilibrium's price, every firm's share, and every firm's markup.

## N.4 Where the match breaks

Now let the regulation arrive: multiply the followers' cost $c_f$ by $1 + \varepsilon > 1$ and hold the calibrated parameters fixed. In the homogeneous model the leader's price rises, the strategic response to its followers' higher cost. In the calibrated system the leader's price stays at $\sigma_U c_L/(\sigma_U - 1)$: an atomistic firm does not respond to a cost shock in the other tier, because followers reach it only through the aggregator. The followers' index $P_F$ rises, the leader's expenditure share rises, and the reallocation runs in the same direction as in the homogeneous model, but not by the same amount. The two structures agree on every fitted moment at the baseline and separate when the regulation arrives.

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